

**UPSC Prelims**

# Environment & Ecology

**Class No. 6**

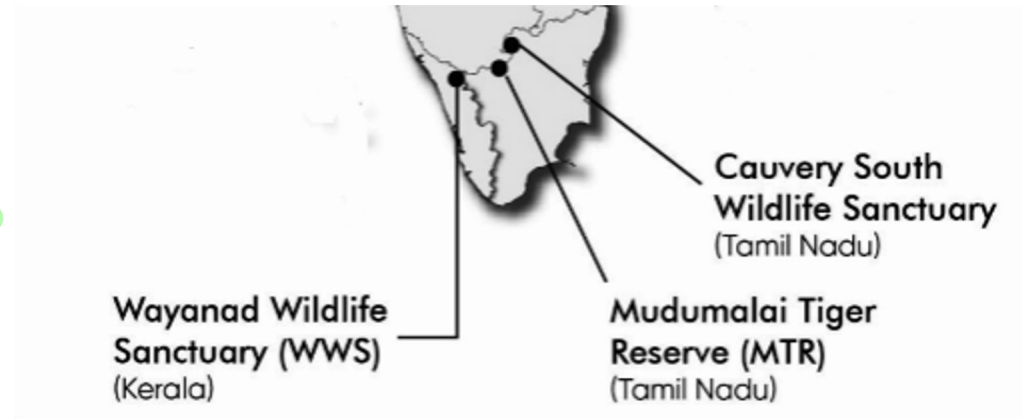
**WLS / TR /ER / Ramsar Sites**

# Wildlife Sanctuary

- **Wayanad Wildlife Sanctuary (WWS)**
- Southern trenches of famous Western Ghats and contiguous to protected areas of Nagarhole and Bandipur National Parks of Karnataka on north-east and Mudumalai National Park of Tamil Nadu on south east.
- **Rivers:** Kabini river (tributary of Cauvery River).
- **Four-horned antelope and Money spider (Recently discovered)** found here.

Ant mimicking spider

Biosphere Reserve → Nilgiri



# Wildlife Sanctuaries

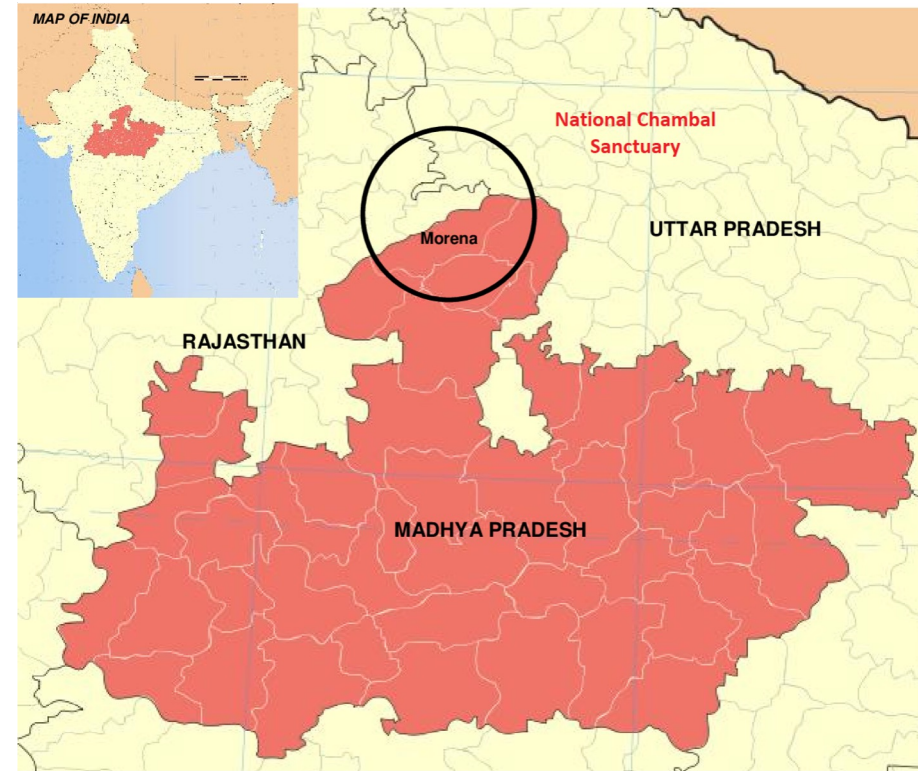
- **Changthang Cold Desert Wildlife Sanctuary**
- First dark sky reserve at Hanle in Ladakh as a part of Changthang Wildlife Sanctuary.
- Situated between Ladakh and Tibet (China) border.
- Home to highest altitude water lakes, namely, Tso Moriri (a Ramsar Wetland site), Pangong Tso and Tso Kar. Indus River flows here.
- **Karakoram (Nubra Shyok) WLS**
- Situated just to the north of Hemis National Park, east of Deosai National Park and to the northwest of the Changthang WLS. Shyok River passes through it.



↳ Mapping concerns.

# Wildlife Sanctuary

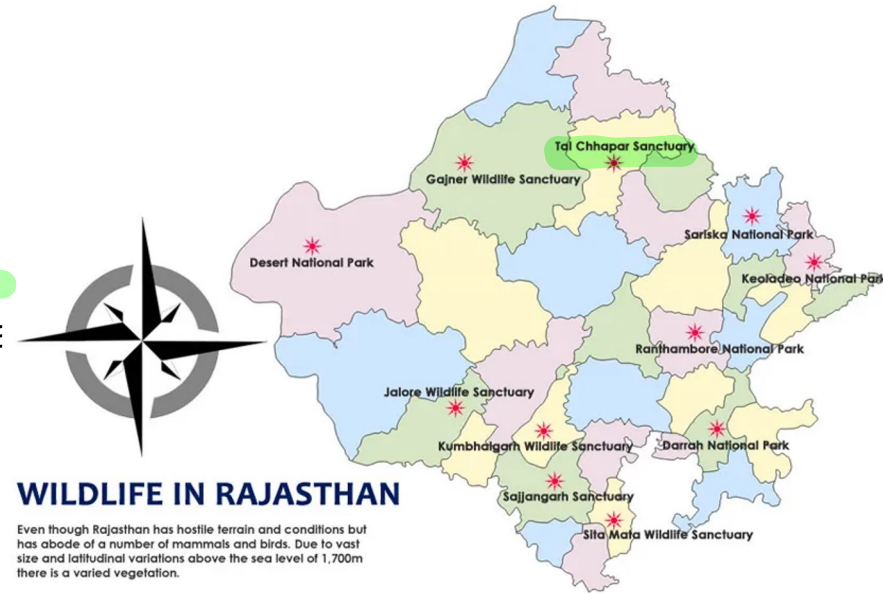
- **National Chambal Sanctuary**
- **Riverine sanctuary**, Species reintroduction programme of **Gharial**
- Listed as an **Important Bird Area (IBA)**.
- Rivers: Chambal River (tributary of Yamuna River) and its tributary Parvati.
- Sand Mining → Legalise



# Wildlife Sanctuary

- **Tal Chhapar Blackbuck Sanctuary**
- Thar Desert, Rajasthan
- Tal is Rajasthani word means plane land.
- Only sanctuary in India with a **good number of blackbucks** in an almost tree-less, saline and flat land.
- **“Moithiya” grass** is found here.

• Rajasthan HC → complete prohibition on any action to reduce area.



# Wildlife Sanctuary

- **Sepahijala Wildlife Sanctuary (Tripura)**
- A woodland with an artificial lake and zoological gardens. (A kind of Zoo plus Garden and Sanctuary.)
- Clouded leopard, Pig-tailed macaque, Barking deer etc found.
- **Cauvery South Wildlife Sanctuary** *act?*
- 17th wildlife sanctuary of Tamil Nadu. *2 elephant corridors*
- Maintains continuity with the Nilgiri Biosphere through Malai Mahadeshwara Wildlife Sanctuary, Billigiri Rangaswamy Temple Tiger Reserve (Karnataka), Sathyamangalam Tiger Reserve and **Cauvery Wildlife Sanctuary in Karnataka.**



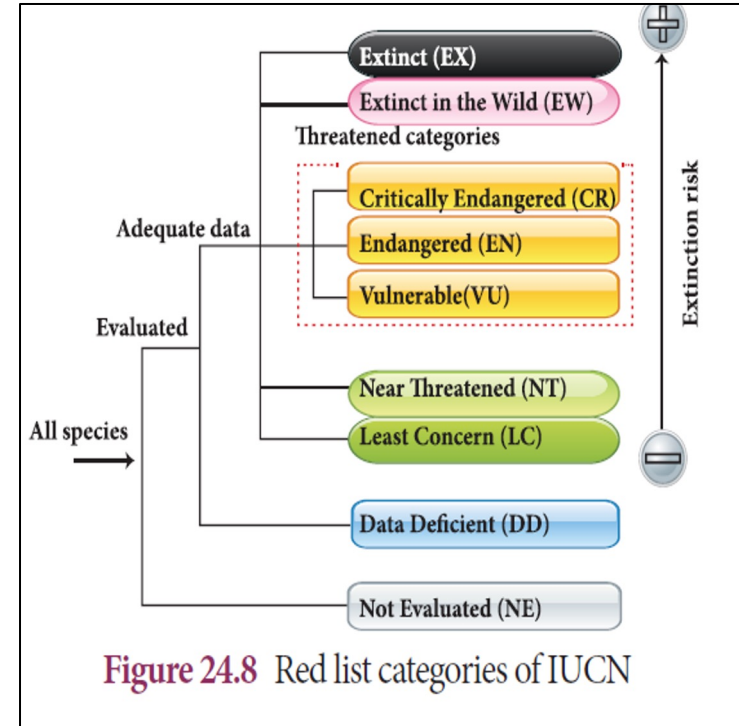
# Conservation Reserves

- **Dugong Conservation Reserve**
- Recently, the Tamil Nadu government has decided to go ahead with the establishment of India's first conservation reserve for the Dugong in Gulf of Mannar, Palk Bay.

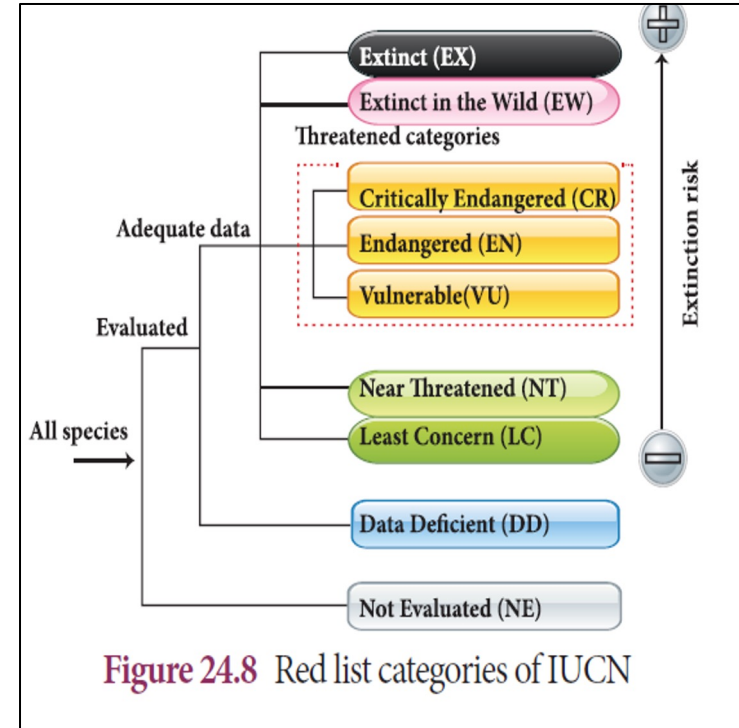


- Was founded in October 1948 as the International Union for the Protection of Nature (or IUPN) following an international conference in Fontainebleau, France.
- Organization changed its name to the International Union for Conservation of Nature and Natural Resources in 1956 with the acronym IUCN (or UICN) with its headquarters in Gland, Switzerland.
- IUCN Members include both States and non-governmental organizations.
- Priority Areas:
  - Biodiversity
  - Climate Change
  - Sustainable Energy
  - Human well-being
  - Green Economy

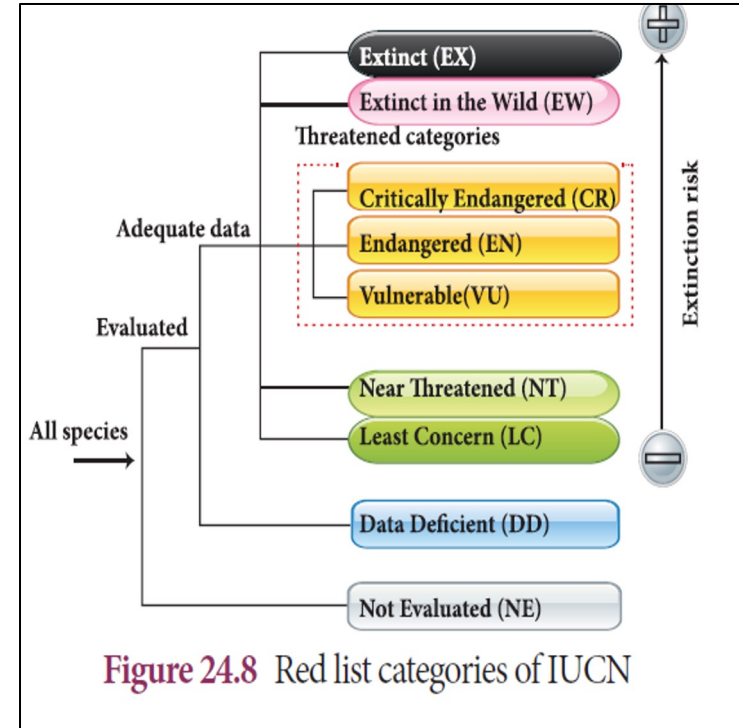
- **Red Data Book-**
- Red Data book or Red list is a catalogue of taxa facing risk of extinction.



- **Critically Endangered Species**
- Reduction in population ( $> 90\%$  over the last 10 years),
- A category containing those species that possess an extremely high risk of extinction as a result of rapid population declines of 80 to more than 90 percent over the previous 10 years (or three generations), a current population size of fewer than 50 individuals, or other factors



- **Endangered (EN) species**
- Designation applied to species that possess a very high risk of extinction as a result of rapid population declines of 50 to more than 70 percent over the previous 10 years (or three generations), a current population size of fewer than 250 individuals, or other factors



- Vulnerable (VU) species
- A category containing those species that possess a very high risk of extinction as a result of rapid population declines of 30 to more than 50 percent over the previous 10 years (or three generations), a current population size of fewer than 1,000 individuals, or other factors
- Near threatened (NT)
- Likely to become endangered soon.

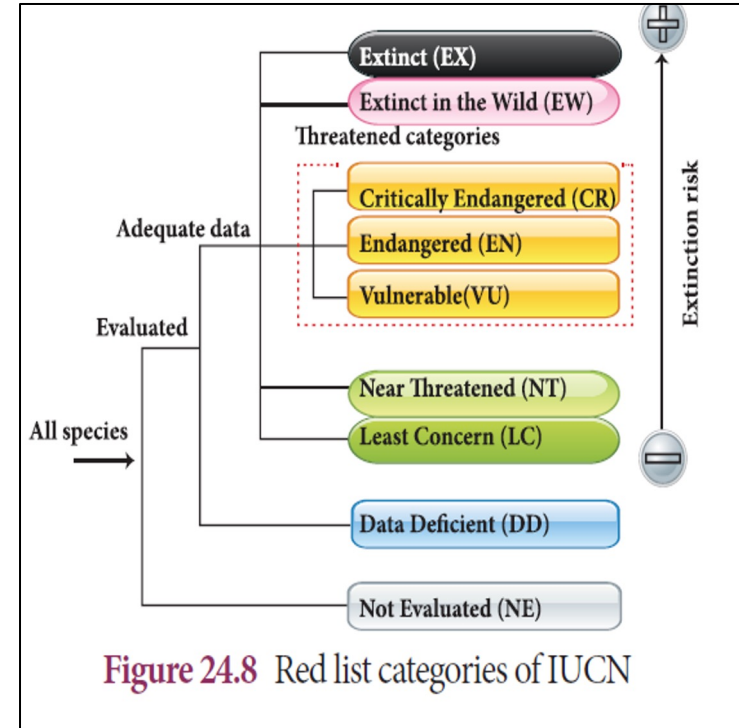
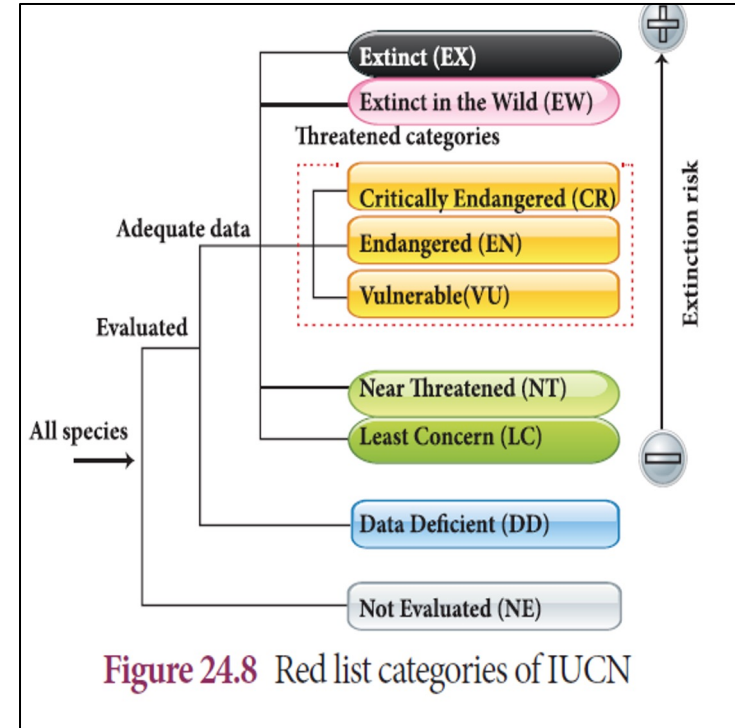


Figure 24.8 Red list categories of IUCN

- **Extinct**
- A species becomes extinct when the last existing member of that species dies. Extinction therefore becomes a certainty when there are no surviving individuals that are able to reproduce and create a new generation.
- **Extinct in the Wild**
- Species listed under this status by IUCN are not known to have any living specimens in the wild, and are maintained only in zoos or other artificial environments.



# Types of Species



- **Keystone Species**
- Keystone species are defined as species that have a massive impact on an ecosystem, to the extent that without their presence, the ecosystem would be dramatically different.
- The preservation and restoration of keystone species is essential for maintaining and/or reestablishing the historic structure and function of the ecosystems they inhabit.
- They are not always the largest or most abundant species in an ecosystem.

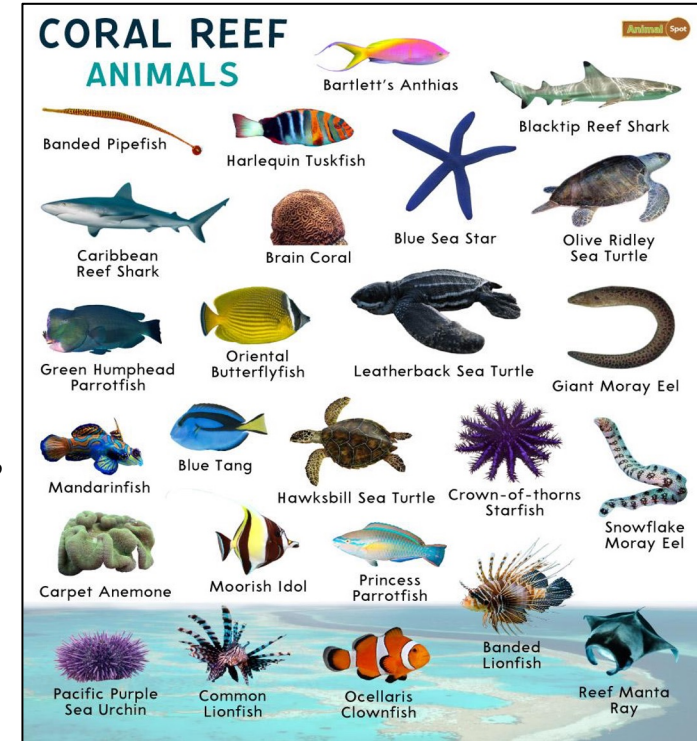
# Types of Species



- **Keystone Species**
- In African savannas such as the Serengeti plains in Tanzania, elephants are a keystone species.
- Elephants eat shrubs and small trees, such as acacia, that grow on the savanna.
- Even if an acacia tree grows to a height of a meter or more, elephants are able to knock it over and uproot it.
- This feeding behavior keeps the savanna a grassland and not a forest or woodland.

# Types of Species

- **Foundation species**
- Foundation species play a major role in creating or maintaining a habitat.
- Corals are a key example of a foundation species
- Coral reefs are one of the most vibrant and biologically diverse ecosystems on the planet.
- Microscopic plankton, as well as crustaceans, mollusks, sponges, fish, and marine reptiles and mammals are all part of healthy coral reef ecosystems.



# Types of Species

- **Indicator species**
- An indicator species describes an organism that is very sensitive to environmental changes in its ecosystem.
- Indicator species are almost immediately affected by changes to the ecosystem and can give early warning that a habitat is suffering.
- Example- lichens (symbiotic relationship between algae and fungi)
- Lichens are sensitive to atmospheric pollution such as nitrogen (N) because they receive all their nutrients and water from wet and dry atmospheric deposition (fall out).
- Nitrogen deposition can increase the load of nutrients. Too much N can harm and kill the algae's chlorophyll which is used to produce sugars feeding it and the fungi.



# Types of Species

- **Flagship Species**
- A flagship species acts as a symbol for an environmental habitat, movement, campaign, or issue.
- They can be mascots for entire ecosystems.
- They are often “charismatic megafauna,”- large animals with popular appeal due to their appearance or cultural significance.
- Example- Giant Panda
- Pandas are the global symbol of endangered species and the value of captive breeding. (WWF logo)



# Tiger

- There are eight subspecies of tiger. Among the eight, at present, five subspecies are present in the wild.
- They are Bengal, South China, Indochinese, Sumatran, and Siberian.
- Three subspecies of tiger — Caspian, Bali, and Javan — are extinct
- **Royal Bengal Tiger (Panthera Tigris)**
- Tiger species native to India.

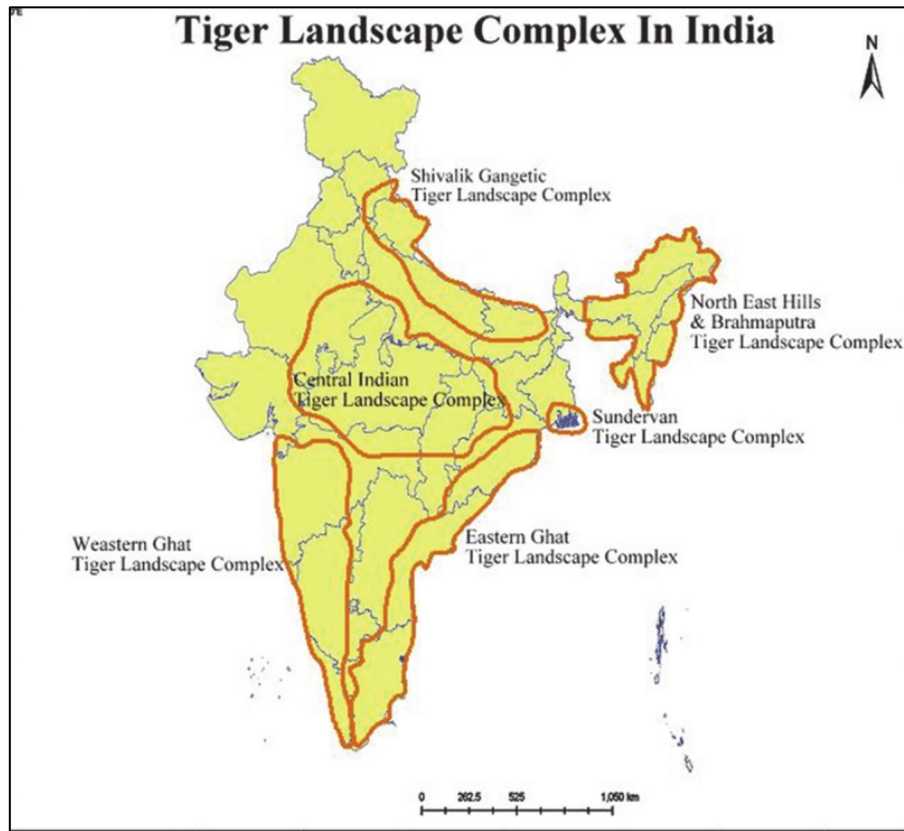


# Tiger

- A **flagship species of India**, declared as the **National Animal of India**.
- **Habitat**: Largest population is in India, smaller groups in **Bangladesh, Nepal, Bhutan, China and Myanmar**.
- India is home to more than **70% of global tiger population**.
- Nearly 35% of tigers in India are found outside tiger reserves
- **Significance of tiger conservation**: Both an **Umbrella species** and a **Keystone species** (has a disproportionately large effect on its natural environment relative to its abundance).



- **Distribution of Tiger population in India:**
- Habitats ranging from the high mountains, mangrove swamps, tall grasslands, to dry and moist deciduous forests, as well as evergreen and shola forest systems.
- **Madhya Pradesh has the maximum number of tigers** followed by Karnataka and Uttarakhand.
- Tiger Census 2018:
- **2967** population
- **3,167** (2022)



# Tiger



- **Characteristics:**
- Solitary and territorial animal, and the territory of an adult male may encompass territories of two to seven females.
- Stripes are individually as unique as human fingerprints
- Strong swimmers
- Home range varies from 15-400 sq.km
- **Home range-** The home range of an animal is the area where it spends its time; it is the region that encompasses all the resources the animal requires to survive and reproduce.

- **Conservation Measures**
- **Project Tiger**
- Project Tiger will complete 50 years in the year 2023.
- Centrally Sponsored Scheme of MoEFCC
- **Implementing Agency:** Statutory body National Tiger Conservation Authority (NTCA), established through Wild Life (Protection) Amendment Act, 2006.

- Project Tiger was launched in Jim Corbett National Park of Uttarakhand, and various tiger reserves were created since then based on a core-buffer strategy.
- **Core area**
- The core areas are notified by the State Government in consultation with an Expert Committee (constituted for that purpose)
- A core area has the legal status of a national park or wildlife sanctuary.
- **Buffer area**
- The buffer zone is the area peripheral to the critical tiger habitat or core area providing supplementary habitat for dispersing tigers, besides offering scope for co-existence of human activity (tribals).

- The limits of such areas are determined with the concerned Gram Sabha and an Expert Committee constituted for the purpose.
- The Forest Rights Act passed by the Indian government in 2006 recognizes the rights of some forest dwelling communities in buffer areas
- **Tiger Census**
- Tiger Census Report is a four-yearly report. It is carried out by the Wildlife Institute of India (WII — autonomous institution under the Ministry of Environment Forest and Climate Change) and the NTCA

- **Methods used for population estimation:**
  - Pugmark Census Technique
  - camera trapping and DNA fingerprinting.
- **MSTrIPES (Monitoring system for Tigers – Intensive Protection and Ecological Status)** was launched across Indian tiger reserves by NTCA and WII in 2010.
- It is a software-based monitoring system, designed to assist wildlife protection.

# Tiger



- 2018 census was the fourth cycle of the tiger census
- The All-India Tiger Estimate 2022 has been published **yesterday**.
- India's five tiger landscapes are:
  - 1. Shivalik Hills and Gangetic Plains,
  - 2. Central Indian Landscape and Eastern Ghats,
  - 3. Western Ghats, North-East Hills and
  - 4. Brahmaputra Plains, and
  - 5. the Sundarbans.

- **2018 census**
- Total population- 2967
- Madhya Pradesh (526) has the highest tiger population
- Karnataka (524) has the second-highest tiger population.
- Uttarakhand (442) has the third-highest tiger population.
- **Central Indian Landscape and Eastern Ghats** had the highest tiger population
- Corbett Tiger Reserve recorded the highest number of tigers (266).
- India achieved the goal of doubling the tiger numbers in 2018, four years before the targeted year 2022.

- **International Measures**
- **The Global Tiger Initiative**
- It was launched in 2008 as a global alliance of governments, NGOs, civil society, and scientific communities with the aim of working together to save wild tigers from extinction.
- In 2013, the scope was broadened to include Snow Leopards
- It is led by the 13 tiger range countries — Bangladesh, Bhutan, Cambodia, China, India, Indonesia, Laos, Malaysia, Myanmar, Nepal, Russia, Thailand, & Vietnam.

- **St. Petersburg Declaration**
- In 2010, leaders of the tiger range countries (TRCs) assembled at an International Tiger Forum in St. Petersburg, Russia, to adopt the St. Petersburg Declaration on Tiger Conservation.
- Their overarching goal was to double the number of wild tigers (TX2) across their geographical area from about 3,200 to more than 7,000 by 2022.
- **TX2 and Tiger Conservation Excellence Award**
- TX2 Award and Tiger Conservation Excellence Award are awarded to sites that are contributing towards achieving the TX2 (Tigers times two) goal — to double the population of wild tigers by 2022.
- Pilibhit Tiger Reserve won the inaugural 2020 TX2 Award.

- **Conservation Assured | Tiger Standards (CA|TS or CATS)**
- CA|TS is a set of criteria which allows tiger sites to check if their management will lead to successful tiger conservation.
- 17 Tiger Reserves in India have CA|TS international accreditation

# Tiger



- **Tiger Reserves in India have CA/TS international accreditation**
- Mudumalai, Anamalai, Sathyamangalam Tiger Reserves Tamil Nadu
- Bandipur Tiger Reserve, Karnataka
- Parambikulam Tiger Reserve, Kerala
- Sunderbans Tiger Reserve, West Bengal
- Dudhwa Tiger Reserve, Uttar Pradesh
- Valmiki Tiger Reserve, Bihar
- Pench Tiger Reserve, Maharashtra
- Satpura, Kanha, Panna, Bandhavgarh and Pench Tiger Reserves, Madhya Pradesh

# Tiger Reserves

- **Total:54** (Ranipur Tiger Reserve)
- **Jim Corbett Tiger Reserve**
- Largest Tiger density in the world.
- River Ramganga, a tributary of river Ganga flows here.



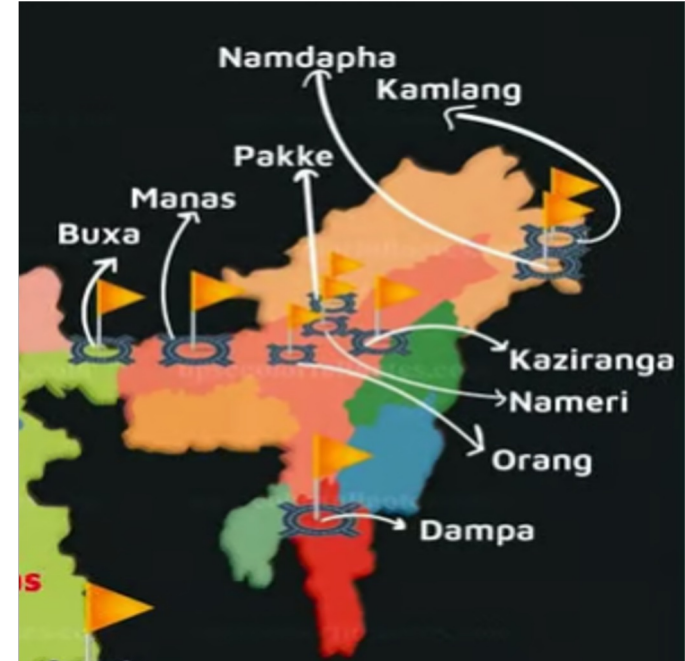
# Tiger Reserves

- **Rajaji, Pilibhit, Dudhwa, Corbett:**  
Second Largest Contiguous Population.
- **Pilibhit TR**
- Forms part of the Terai Arc Landscape
- River Sharda and Ghaghara

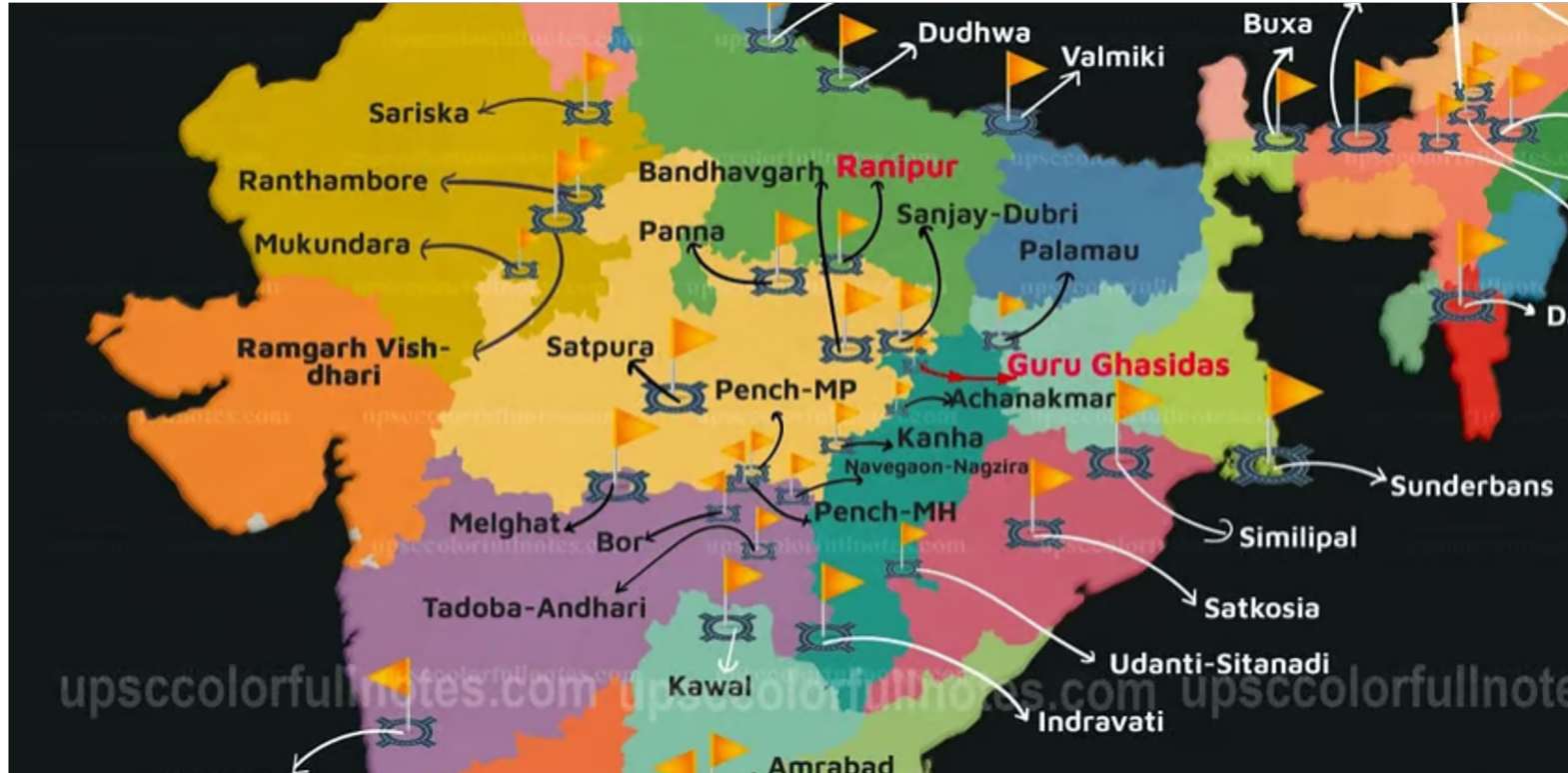


# Tiger Reserves

- **Pakke (Pakhui) TR**
- It lies in the foothills of the Eastern Himalaya.
- Rivers: Kameng River (west) and north, and by Pakke River (east).
- To the south, the sanctuary adjoins Assam's Nameri NP and to the west, it is bounded by Eaglenest WLS and Sessa Orchid Sanctuary.
- It is known for its **Hornbill Nest Adoption Programme**.
- The Hornbill Nest Adoption Program (HNAP) is a community-based conservation initiative that was started in 2012 to protect hornbills and hornbill nest trees/habitat. The program is based on the concept of “shared parenting”:



# Tiger Reserves



# Recently Declared Tiger Reserves



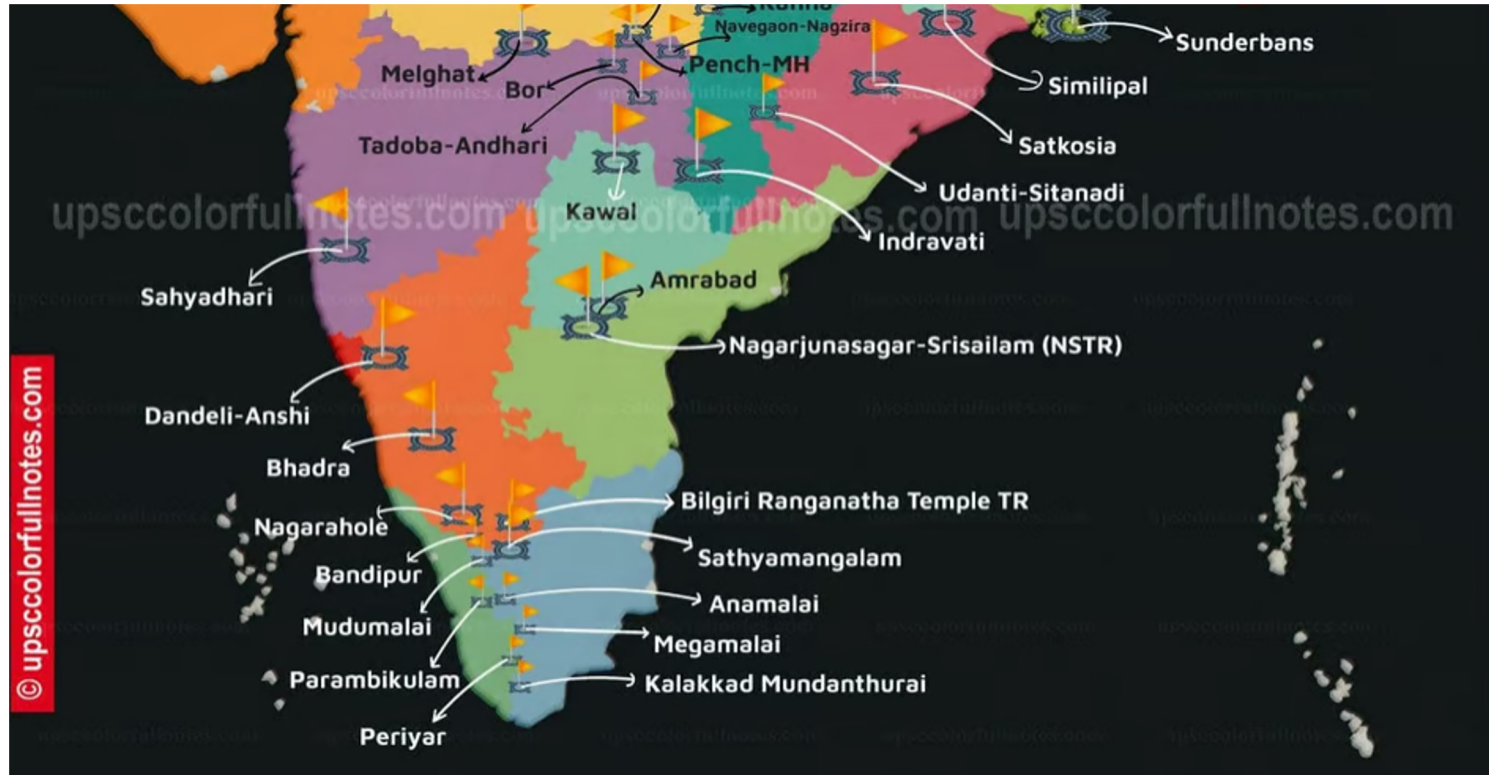
- **Ramgarh Vishdhari Tiger Reserve**
- Rajasthan Government has notified the Ramgarh Vishdhari Wildlife Sanctuary as a tiger reserve as 4th Tiger Reserve of Rajasthan. → *Ranthambore, Sariska, Mukundra*
- It serves as a tiger corridor between Ranthambore Tiger Reserve and Mukundra Hills Tiger Reserve. River: Mez, a tributary of Chambal river.
- **Ranipur Tiger Reserve**
- India's 53rd tiger reserve in Ranipur Wildlife Sanctuary in **UP**.
  - *Dudhwa*
  - *Pilibhit*
  - *Amangesh.*
- **Sariska Tiger Reserve**
- A Tiger was recently shifted from **Ranthambore Tiger Reserve (RTR) to Sariska Tiger Reserve (STR)**.
  - *Chambal/Banas river*
  - *Ranthambore fort*
- Part of the Aravalli Range.

# Recently Declared Tiger Reserves



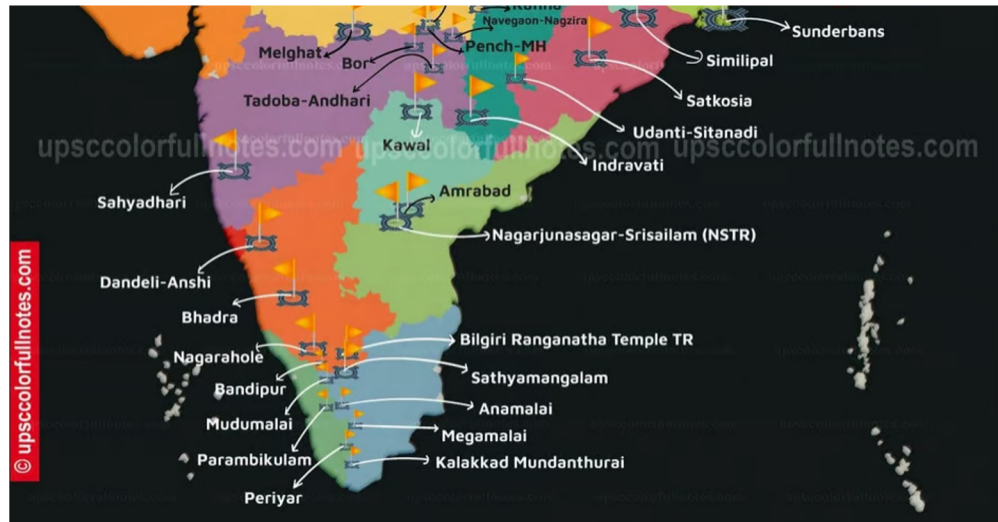
- **Durgavati WLS**
- Madhya Pradesh (MP) Wildlife Board **approved a new reserve for tigers of Panna Tiger Reserve (PTR).**
- **One-fourth of PTR will get submerged due to linking of Ken-Betwa rivers.**
- In wake of river linking project, NTCA had asked UP and MP governments to **notify new tiger reserves.**
- The proposal will be sent to **National Tiger Conservation Authority (NTCA)** for final approval.
- Durgavati Wildlife Sanctuary is a **green corridor for Bandhavgarh Tiger Reserve.**

# Tiger Reserves

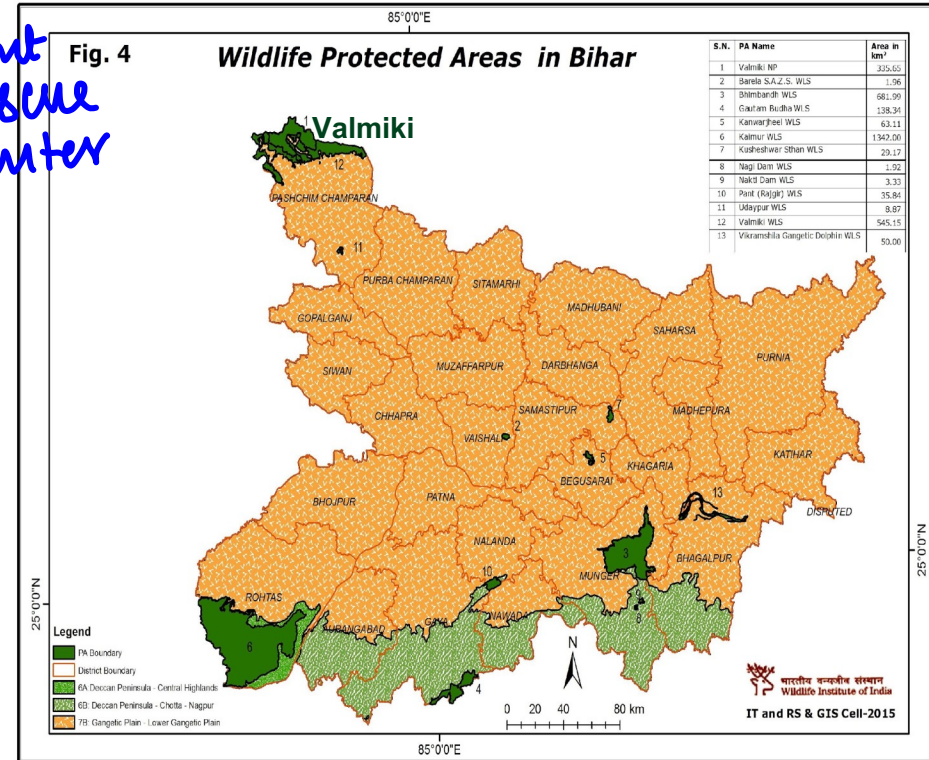


# Tiger Reserves

- **Tamil Nadu and Kerala:** Largest Contiguous Population of Tigers
- **Nagarjunasagar-Srisailem TR:** Largest contiguous Critical Tiger Habitat
- **Simplipal TR:** Decline in number
- **Satkosia Tiger Reserve**
- Mahanadi Gorge
- Indian Giant Squirrel, Indian Python
- **Mixed Deciduous Sal**

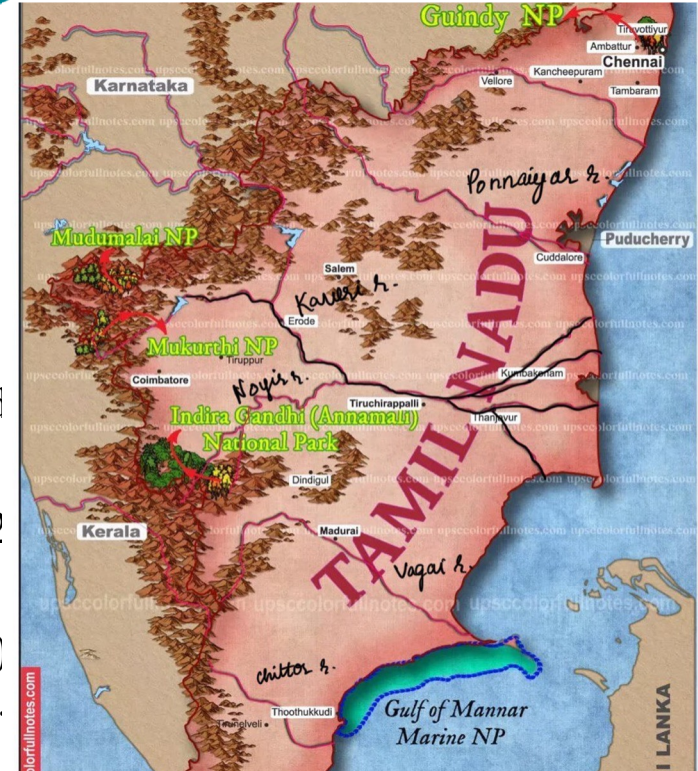


- Valmiki National Park (TR) → elephant rescue center
- Location: Along the international border with Nepal and contiguous with Nepal's Chitwan National Park to the north
- Only protected area where Gaur is found in the Shivaliks and Gangetic plains landscape
- Among the 14 tiger reserves in India which received accreditation of Global Conservation Assured Tiger Standards (CA|TS).
- Rivers: Gandak River on the West boundary



# Tamil Nadu

- **Guindy National Park**
- Tropical dry evergreen forest, Sapta Kannika Grove
- Gharial Captive Breeding
- **Mudumalai National Park (TR)**
- Endemic fish species namely, Puntius mudumalaiensis and White Backed Vultures and Long Billed Vulture (CR)
- Newly captured and reintroduced wild elephant PM2 (Pandalur Makhna) in MTR is adapting to new habitat.
- Shares boundary with Wayanad Wildlife Sanctuary (Kerala) on West, Bandipur Tiger Reserve (Karnataka) on North. Part of the Nilgiris Biosphere Reserve. Moyar river.
- Among the 14 tiger reserves in India under CA/TS.



# Elephant

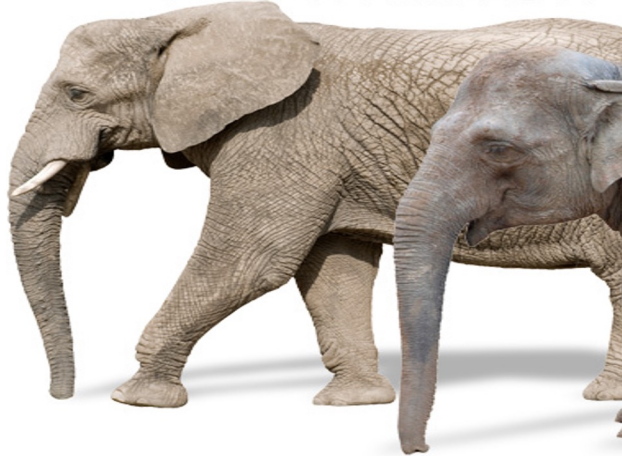
- **Indian elephant (*Elephas maximus*)**
- **Habitat:** Found in the central and southern Western Ghats, North East India, eastern India and northern India and in some parts of southern peninsular India.
- As per the last count in 2017, India has 29,964 elephants.
- Karnataka has the highest number of elephants, followed by Assam and Kerala.



- **Characteristics:**
- Highly intelligent animals characterised by strong family bonds, sophisticated forms of communication and complex behaviour, including tool use and the ability to feel grief and compassion.
- Form herds of related females led by the oldest female, the 'matriarch'.
- Have longest gestation period of all mammals (18 to 22 months).
- Adult male Asian elephants are less social than females. They enter musth -- a mate-searching strategy for old (above 30 years of age) males, annually.

# Elephant

## AFRICAN ELEPHANT



**EARS:** Large, "Africa-shaped" ear  
**TRUNK TIP:** Two "fingers" for grasping  
**HEAD:** Single dome  
**HEIGHT** (*tallest at shoulder*): 9 – 13'  
**WEIGHT:** 8,800 – 15,400 lbs.

## ASIAN ELEPHANT

*What are the differences?*



**EARS:** Small, rounded ear  
**TRUNK TIP:** One "finger" for grasping  
**HEAD:** Twin dome  
**HEIGHT** (*tallest at back*): 6.5 – 9'  
**WEIGHT:** 6,600 – 13,200 lbs.

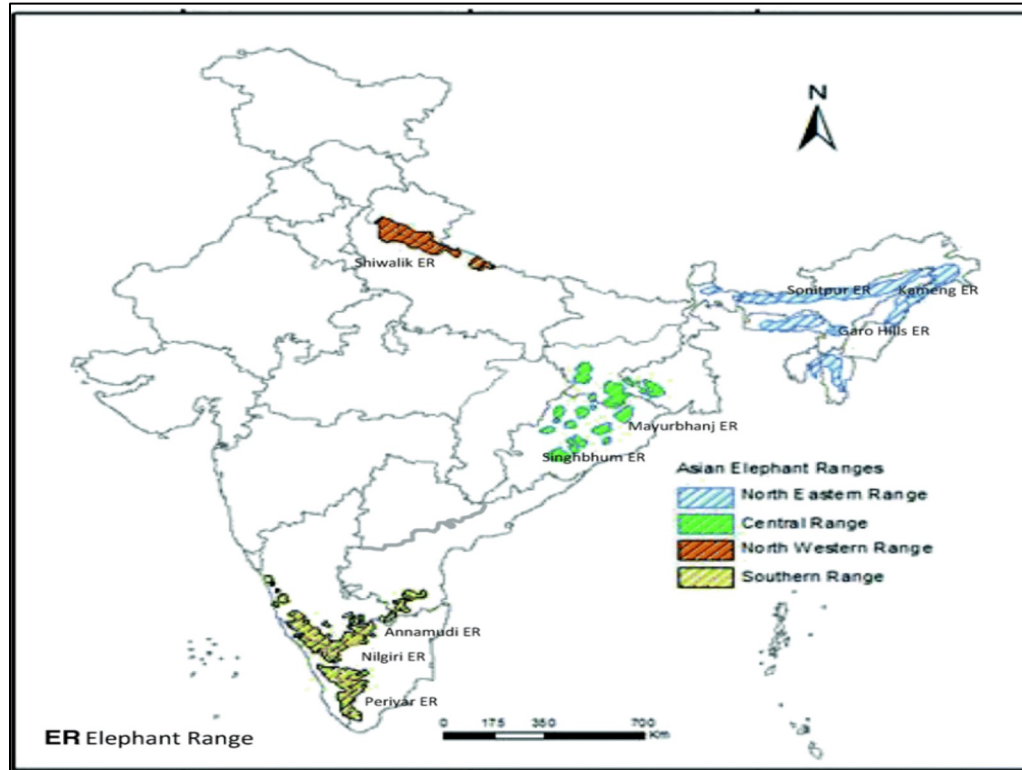


# Elephant



- **Elephant reserves in India**
- There are 33 notified Elephant Reserves (ERs) in the country spread over 15 states.
- Tamil Nadu and Assam have the highest number of ERs (5 each) followed by Kerala (4) and Odisha (3).
- Mysore ER in Karnataka is the largest ER.
- **Elephant Corridors : no legal recognition.**
- Elephant corridors are narrow strips of land that connect two large habitats of elephants.
- Elephant corridors are crucial to reduce animal fatalities due to accidents and other reasons

# Elephant



# Elephant



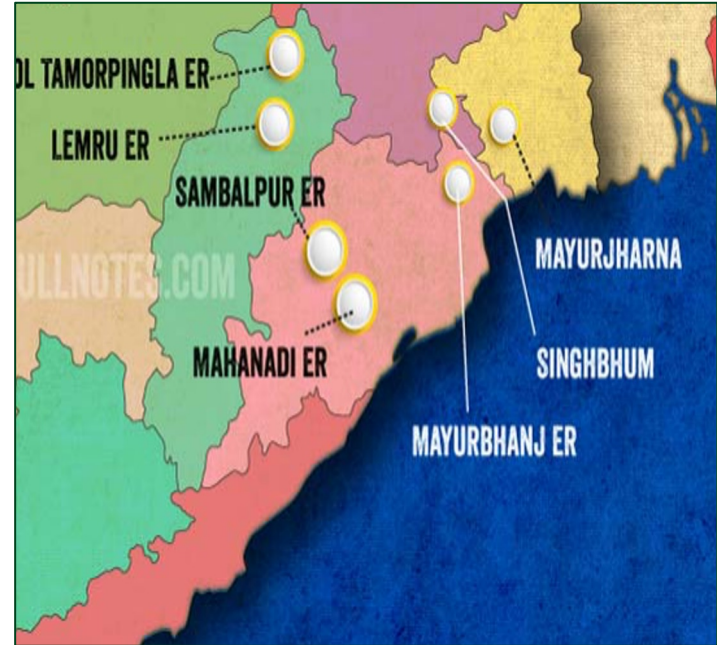
- **Threats:** Man-elephant conflict, habitat loss and fragmentations, poaching and the loss of genetic viability resulting from small population size and isolation.
- **Conservation measures**
- India launched '**Project Elephant**' in 1992. Under the project, the government provides technical and financial help to states to save elephants.
- Development of Elephant Reserves.
- **Monitoring the Illegal Killing of Elephants (MIKE)**, an international effort for conservation of elephants in Asia and Africa

- **Project RE-HAB**
- RE-HAB Project is an innovative way of preventing elephant–human conflicts without causing any harm
- Under this project, bee boxes are used as a fence to prevent elephants from entering human habitation.
- Elephants fear that the honeybees might sting them in their eyes and the inner side of the trunk.
- The buzz of the bees irritates the elephants the most
- It is demonstrated by Khadi and Village Industries Commission's (KVIC)

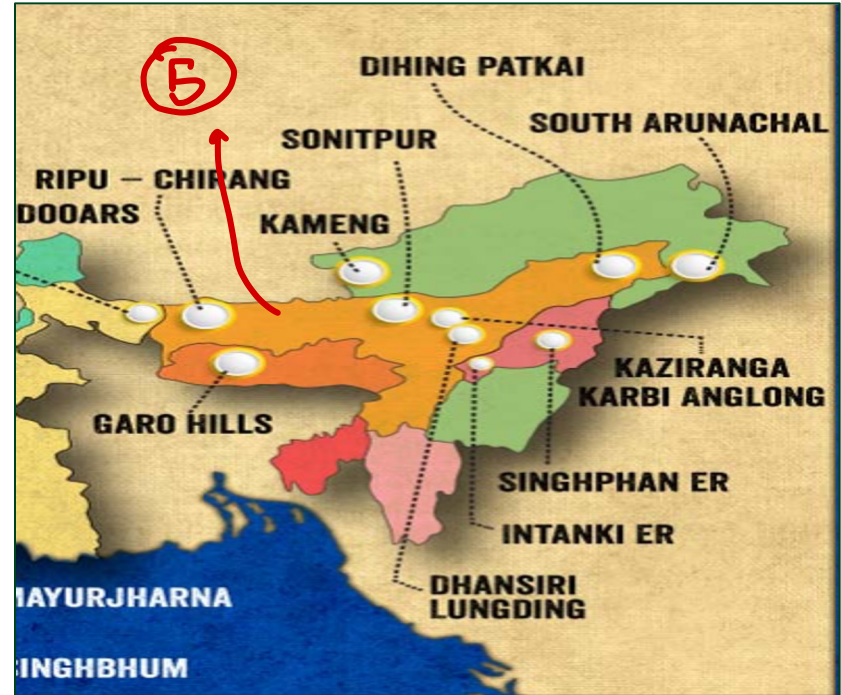
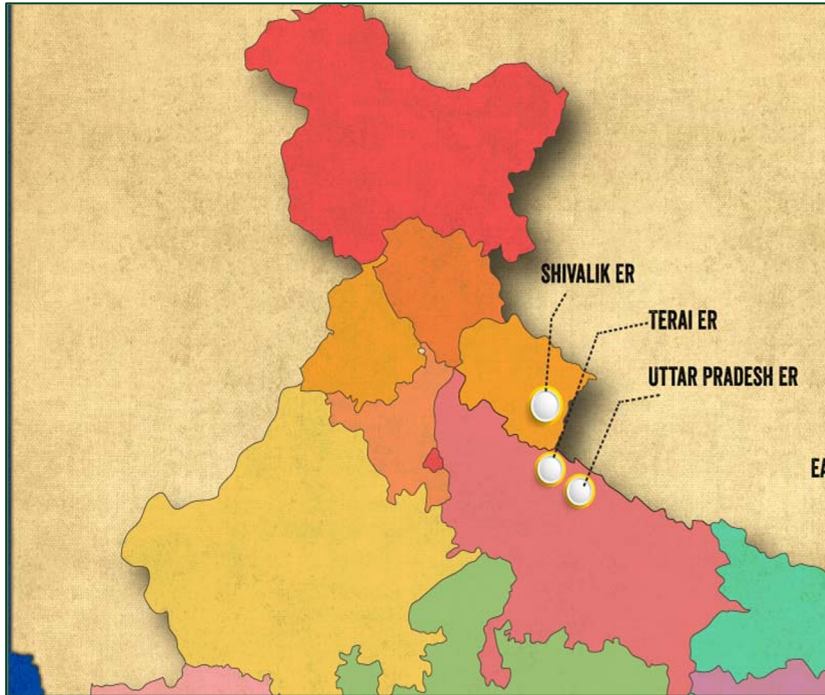
- **Project Elephant**
- Tiger faces threat of extinction, whereas the elephant faces threat of attrition.
- The elephant numbers have not increased or decreased drastically but there is an increasing pressure on the elephant habitats.
- Project Elephant was launched in 1992.
- It is a centrally sponsored scheme.
- **Elephant corridor**
- A stretch/narrow strips of forested (or otherwise) land that connects larger habitats with elephant populations and forms a conduit for animal movement between the habitats.

# Elephant Reserves in Southern and Central India

101 elephant corridors have been identified by the Wildlife Trust of India under the National elephant Corridor project.



# Elephant Reserves in Northern and Eastern India



# Latest Elephant Reserves



- **Terai Elephant Reserve**
- UP, to be established in Dudhwa and Pilibhit tiger reserves
- **Lemru Elephant Reserve**
- In Chhattisgarh's Korba district. The reserve area is part of the Hasdeo Aranya forests, a complex ecosystem rich in coal resources
- **Agasthiyamalai Elephant reserve**
- Tamil Nadu
- It has the potential to link populations in the Srivilliputhur Meghamalai Tiger Reserve and the Periyar ecosystems.

- **Shivalik Elephant Reserve**
- Uttarakhand govt announced **repeal of the decision of denotifying the Shivalik Elephant Reserve** to expand the airport in Dehradun.
- Much of this reserve falls in **Rajaji National Park, Corbett Tiger Reserve, Nanda Wildlife Sanctuary, and Sonanadi Wildlife Sanctuary.**
- It is the home to almost all of Uttarakhand's wild elephants.
- **Monitoring the Illegal Killing of Elephants (MIKE) site**

# Ramsar Sites



- When a country accedes to the Ramsar Convention, it must designate at least one wetland as a Wetland of International Importance. Including a “Ramsar Site” in the list embodies the government’s commitment to take the steps necessary to ensure that its ecological character is maintained. (Criteria and Ramsar Convention has been dealt earlier)
- Today, the Ramsar List is the world’s largest protected area network. Over 2,400 Ramsar Sites cover 2.5 million sq km on the territories of 171 Ramsar Contracting Parties worldwide.
- The world’s first site was **the Cobourg Peninsula in Australia, designated in 1974.**
- The countries with the most Ramsar Sites are the United Kingdom with 175 and Mexico with 142.
- **Bolivia has the largest area under Ramsar protection.**



# Ramsar convention



**Genesis:** An **intergovernmental treaty** adopted in 1971 (came into force in 1975)



**Objective:** Provide the framework for the conservation and wise use of wetlands and their resources.

- **Wise use of wetlands** is defined as the maintenance of their ecological character, achieved through the implementation of ecosystem approaches, within the context of sustainable development.



**Membership:** 172 contracting parties



**Other key information:**

- Has 6 formally recognized International Organization Partners:
  - ➔ BirdLife International
  - ➔ International Union for Conservation of Nature (IUCN)
  - ➔ International Water Management Institute (IWMI)
  - ➔ Wetlands International
  - ➔ WWF International
  - ➔ Wildfowl & Wetlands Trust (WWT)
- Ramsar Sites which are potentially at risk as a result of technological developments, pollution or other human interference may be placed on **Montreux Record**.

- COP : conference of parties to the Ramsar convention : meets every 3 years. COP 14 took place in wuhan, china in November 2022.
- **Wuhan Declaration and 2025-2030 Global Strategic Framework for Wetlands Conservation** were adopted at COP 14.
- **A vital** commitment of the Contracting Parties is to identify and place suitable wetlands onto the List of Wetlands of International Importance, also known as the Ramsar List.
- Contracting Parties are expected (not mandated) to manage their Ramsar Sites to maintain their ecological character and retain their essential functions and values for future generations.

# Deletion from the List

- A Contracting Party may, because of its urgent national interest, delete or restrict the boundaries of wetlands already included in the list. However, such deletions or restrictions should be compensated for by creating additional nature reserves in the same area or elsewhere.
- No Ramsar Site has ever been “deleted” in this way, and Parties have only extremely rarely restricted the boundaries of a Site on this basis.



# Transboundary Ramsar Sites



- Many important wetlands extend as one ecologically coherent whole across national borders.
- In these cases, COP can agree to establish Ramsar Sites on their territory as parts of a bigger Transboundary Ramsar Site. The authorities on both sides of the border agree to collaborate in managing the site.
- None in India.
- Largest is Congo Basin Ramsar Site.

# The Montreux Record

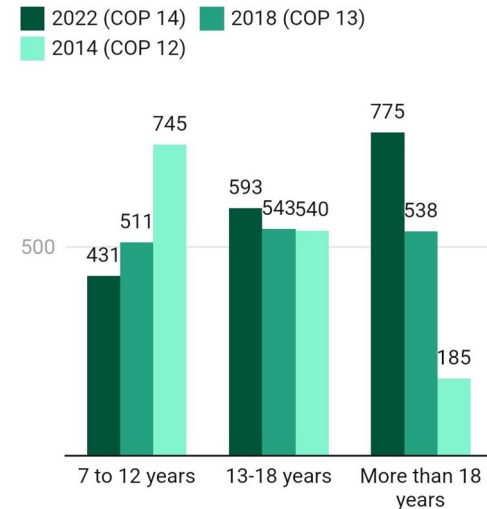
- Ramsar Sites which are potentially at risk because of technological developments, pollution or other human interference may be placed on the **Montreux Record of Ramsar Sites** where changes in ecological character have occurred, are occurring, or are likely to occur.
- Currently, two wetlands of India are in Montreux record: **Keoladeo National Park (Rajasthan)** and **Loktak Lake (Manipur)** (Remember Phumdi).
- Chilika lake (Odisha) was placed in the record but was later removed from it.



# COP14 to the Ramsar Convention

- The event was co-hosted by Wuhan, China and Geneva, Switzerland.
- **14th Conference of the Parties (COP14)** to Ramsar Convention on Wetlands adopted the Wuhan Declaration
- Also, another report presented at COP14 (Wetland Conference) stated that information on 75% of wetlands included in list of Ramsar Sites is outdated.
- Parties to Ramsar Convention are required to update information for their Ramsar Sites at least once every six years.

## Number of Ramsar sites with outdated Information sheets prior to COP14 , COP13 and COP12

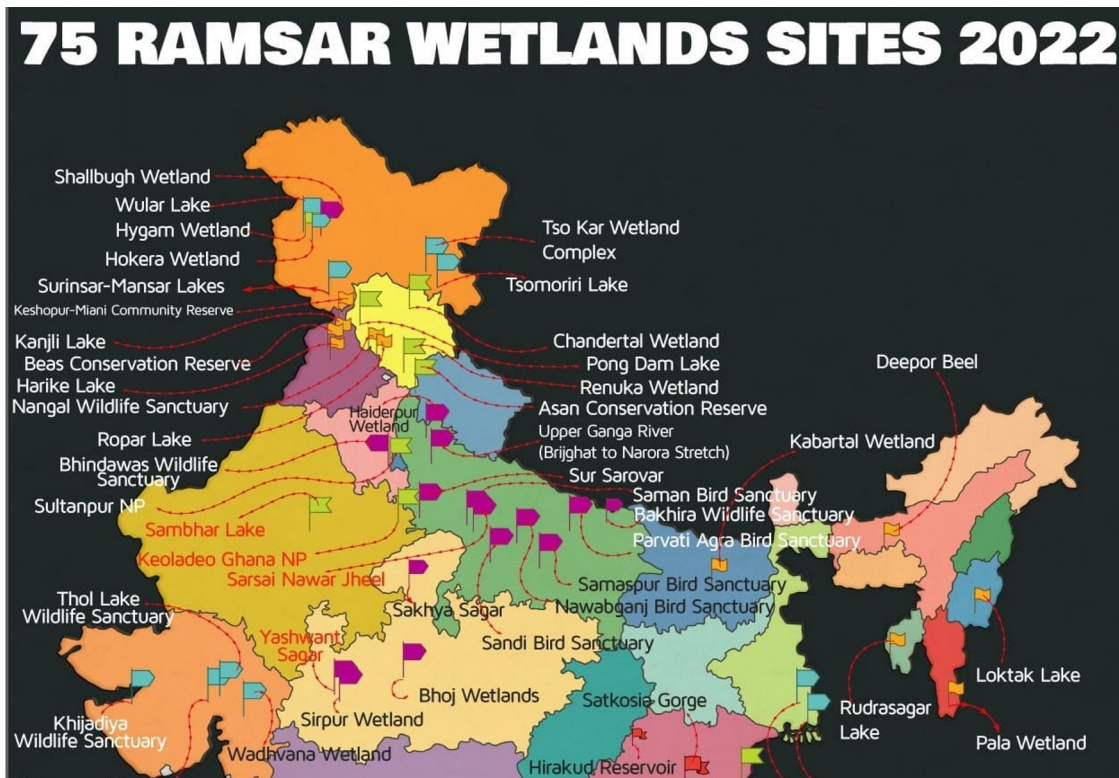


# Wuhan Declaration

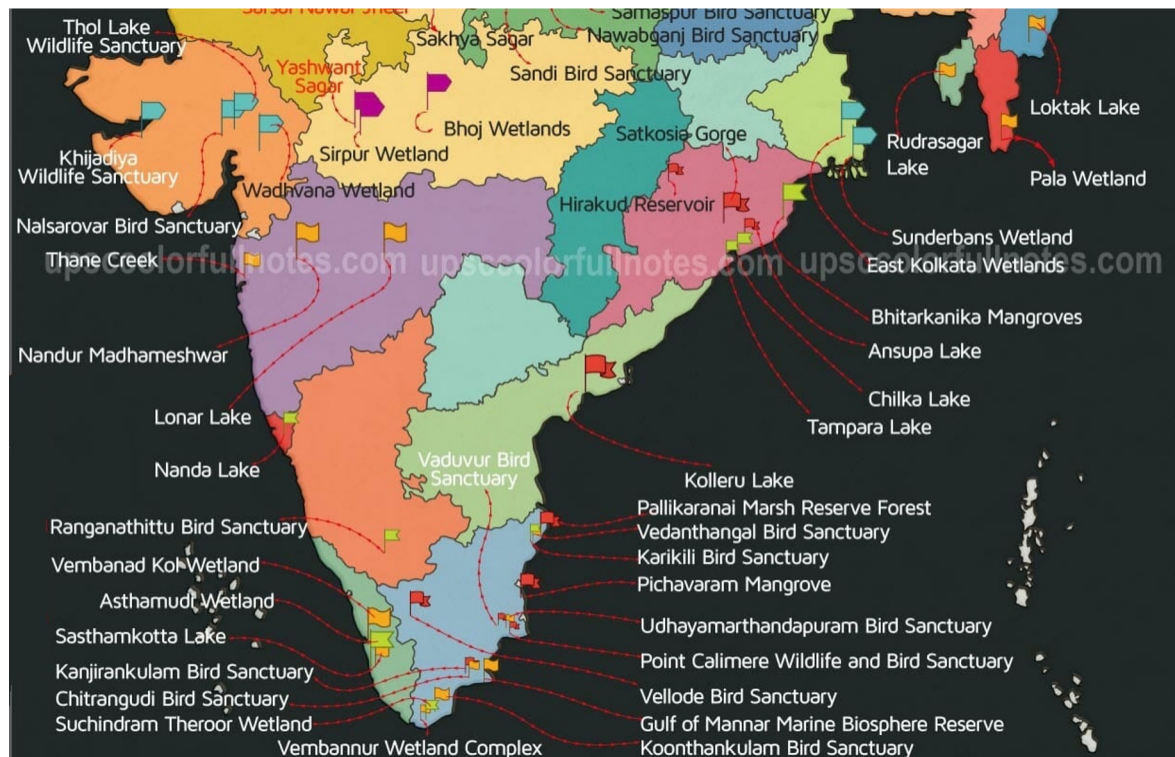


- Wuhan Declaration calls for practical actions to promote conservation, restoration, management and sustainable use of wetlands worldwide.
- Key priorities of Wuhan declaration are:
  - Take measures to halt and reverse wetland loss globally.
  - Incorporate wetlands into national actions to contribute to domestic legislation, plans and implementation of SDGs.
  - Define strategic goals and priority areas for wetlands in partnership with civil society stakeholders, academia and private sector.
  - Importance of healthy wetland ecosystems as nature-based solutions for climate mitigation, adaptation and disaster risk reduction.

# List of Ramsar Sites in India



# List of Ramsar Sites in India



## Chilika lake

- As per the reports, Number of Irrawaddy dolphins in Chilika lake has fallen.
- Chilika Lake, located in east coast of Odisha, is Asia's largest brackishwater lake and second largest coastal lagoon in the world.
- The large Nalabana Island (Forest of Reeds) covering about 16 sq km in the Chilika Lake area was declared a bird sanctuary in 1987.
- **Kalijai Temple:** Located on an island in the Chilika Lake.

## Sunderbans

- It is located within the largest mangrove forest in the world, in the delta of the Rivers Ganges and Brahmaputra.
- It is home to Tigers (Sundarban Tiger Reserve) and a number of other rare and threatened species such as Irrawaddy Dolphin, fishing cat etc.
- Sundarban Wetland, India was recognised as the 'Wetland of International Importance' under the Ramsar Convention in January 2019.

## Sunderbans

- CAG has pointed out delay in coastal zone management plan and construction in sunderbans without CRZ norms being followed.
- CRZ notification designated sunderbans as Critically vulnerable coastal area and any development plan requires permission of WB State coastal zone management authority.
- However, no permission was taken

## East Kolkata Wetlands (EKW)

- Recently, Comptroller and Auditor General pointed out lack of effective action by the East Kolkata Wetland Management Authority.
- East Kolkata Wetlands (EKW) is a unique peri-urban ecosystem on the eastern fringes of Kolkata.
- It is a multiple use wetland with facilities to treat the city's wastewater and use it to support pisci-culture and agriculture.

## Loktak Lake

- The Central Bureau of Communication is organizing a floating photo exhibition at Loktak
- Largest freshwater lake of Northeast India.
- Hosts hundreds of floating islands called Phumdis; Hosts Sangai.
- Other Species of Loktak Lake: Hog Deer, Otter, a host of waterfowls and migratory birds

## Kolleru Lake

- It is located between the deltas of the Krishna and Godavari rivers in Andhra Pradesh.
- It was notified as a Wildlife Sanctuary in 1999 under India's WPA, 1972.
- Wetland of international importance in 2002 under Ramsar convention.
- Atapaka Bird Sanctuary is at Kolleru Lake

## Harike wetland:

- The Harike wetland, one of the largest in northern India, is situated in the Punjab state in India.
- It stands on the confluence of the Beas and the Sutlej.

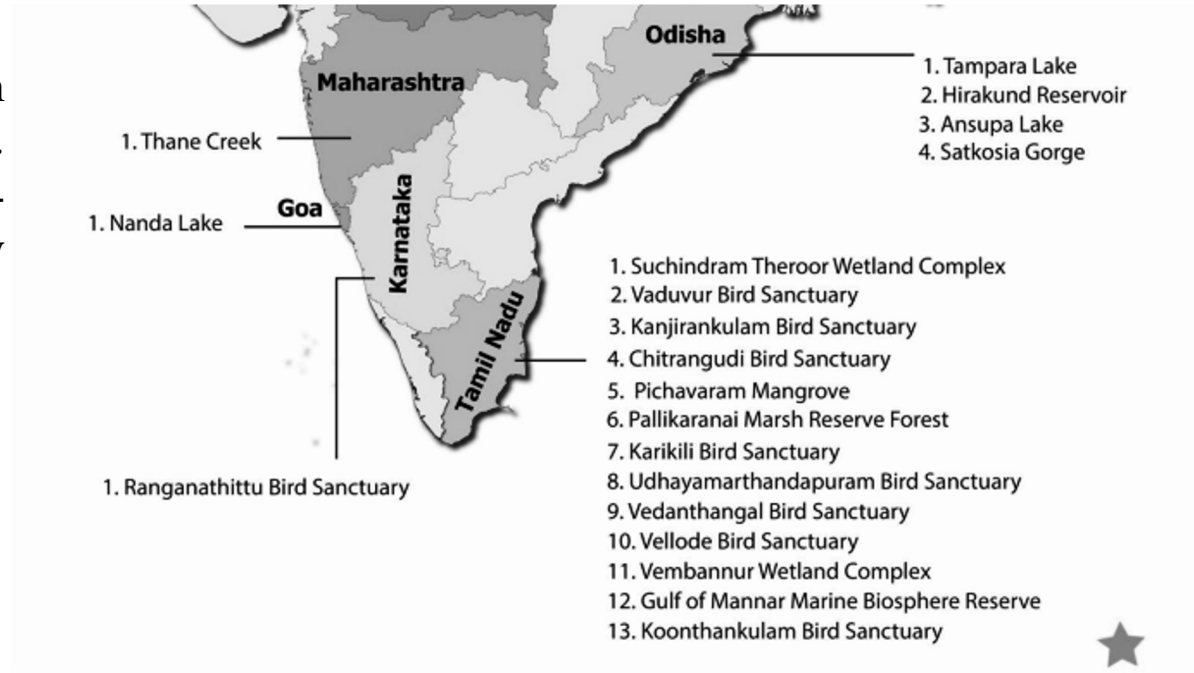
## Keoladeo National Park (Montreux record)

- It is a wetland and bird sanctuary located in Bharatpur, Rajasthan.
- Gambhir (tributary of Yamuna) and Banganga (tributary of Gambhir) flows here.

# Recently Added Ramsar Sites (Tamil Nadu)

## Tamil Nadu

- Tamil Nadu has a maximum no. of Ramsar sites in India. Tamil Nadu has total 14 Ramsar Sites. (13 new additions)
- **Natural wetlands:**
  - Pallikaranai Marsh Reserve Forest
  - Pichavaram Mangrove
  - Gulf of Mannar Marine Biosphere Reserve



# Recently Added Ramsar Sites (Tamil Nadu)



1. Suchindram Theroor Wetland Complex
2. Vaduvur Bird Sanctuary
3. Kanjirankulam Bird Sanctuary
4. Chitrangudi Bird Sanctuary
5. Pichavaram Mangrove
6. Pallikaranai Marsh Reserve Forest
7. Karikili Bird Sanctuary
8. Udhayamarthandapuram Bird Sanctuary
9. Vedanthangal Bird Sanctuary
10. Vellode Bird Sanctuary
11. Vembannur Wetland Complex
12. Gulf of Mannar Marine Biosphere Reserve
13. Koonthankulam Bird Sanctuary

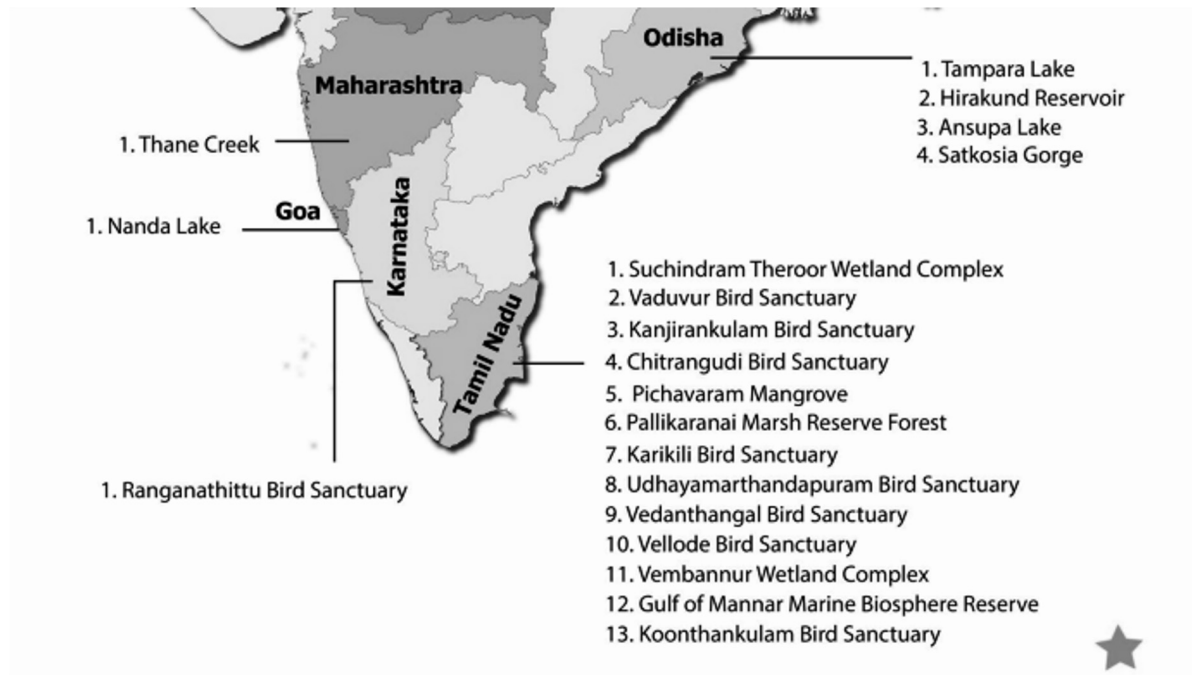
# Recently Added Ramsar Sites

## Ranganathittu Bird Sanctuary (Karnataka)

- Natural wetland
- Part of the Kaveri River.
- Classified as an Important Bird and Biodiversity Area (IBA).

## Nanda Lake (Goa)

- Part of catchment area of Zuari River



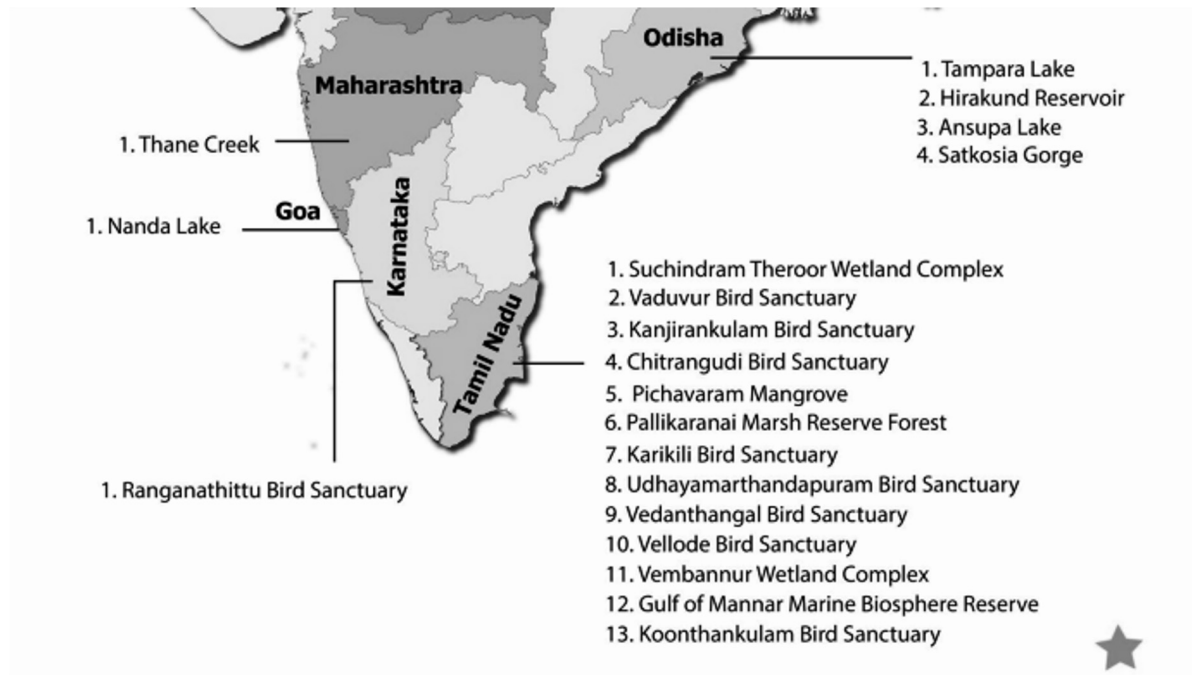
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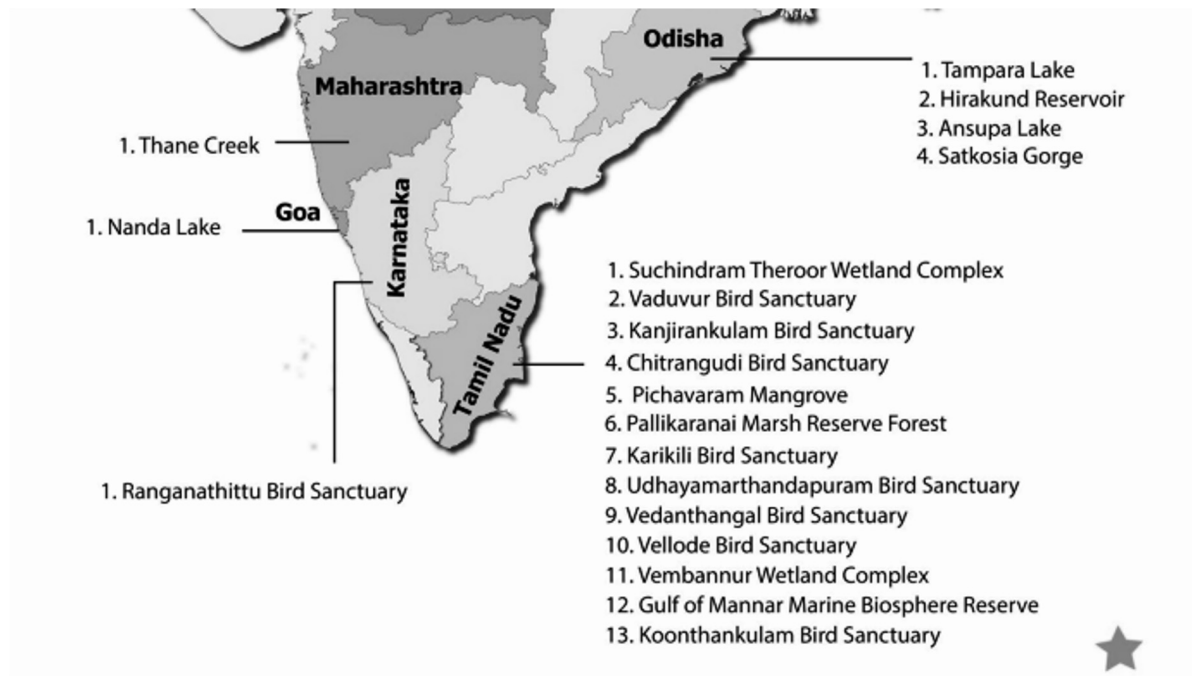
- Part of catchment area of Zuari River



# Recently Added Ramsar Sites (Maharashtra)

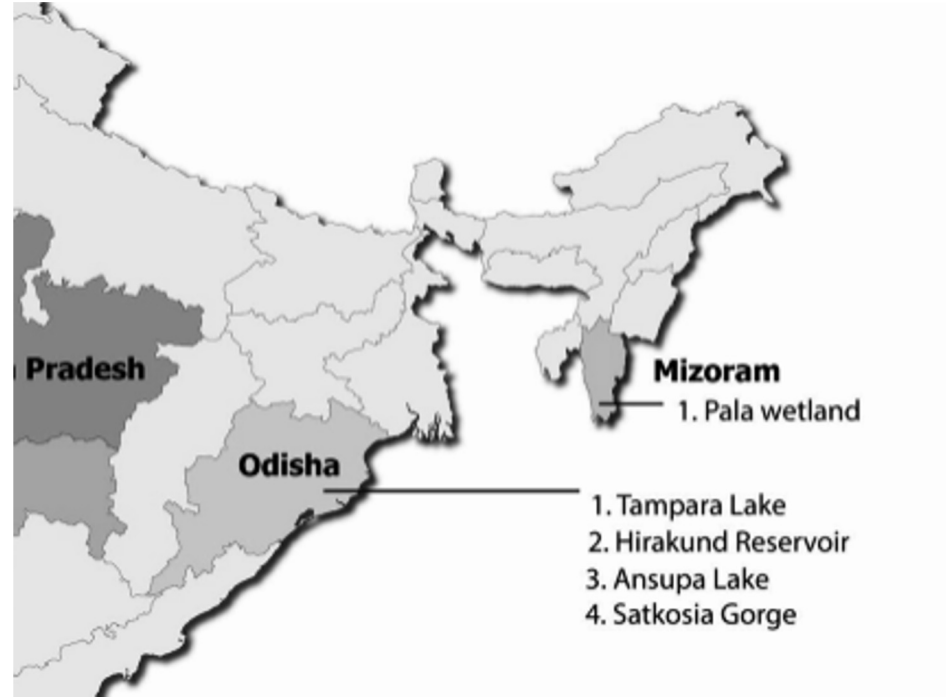
## Thane Creek

- Declared as Thane Creek Flamingo Sanctuary.
- Important Bird Area.
- Ulhas River flows here.



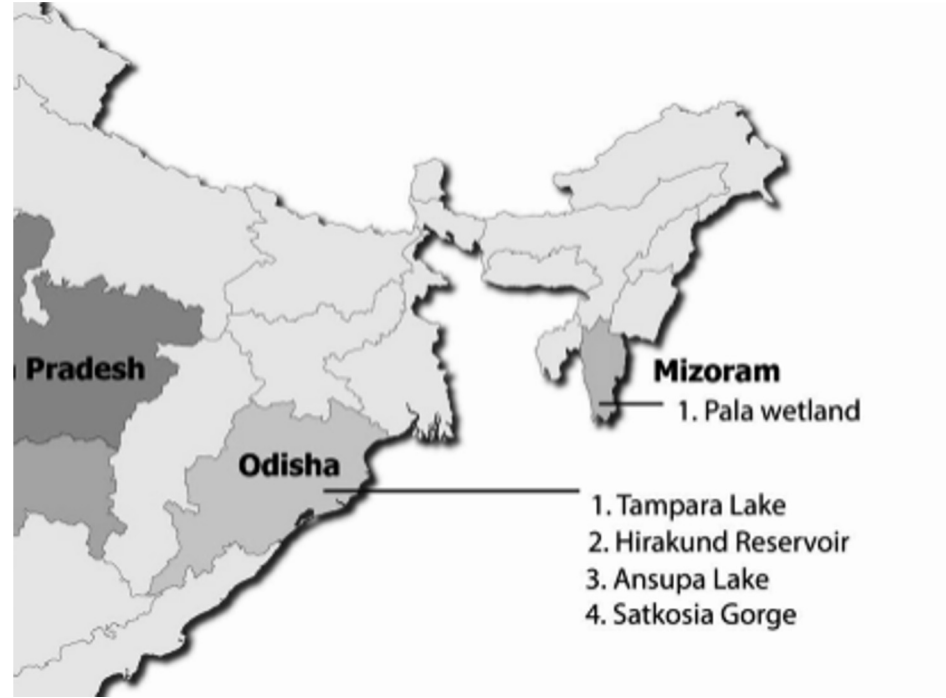
# Recently Added Ramsar Sites (Odisha)

- **Satkosia Gorge (Natural wetland)**
- Established as a wildlife sanctuary and Satkosia Tiger Reserve
- Comprising two adjoining wildlife sanctuaries, the Satkosia Gorge sanctuary and Baisipalli sanctuary.
- Spreads along the gorge over the river Mahanadi.
- Mosaic of rivers, marshes and evergreen forests at the meeting point the Deccan Peninsula and the Eastern Ghats.



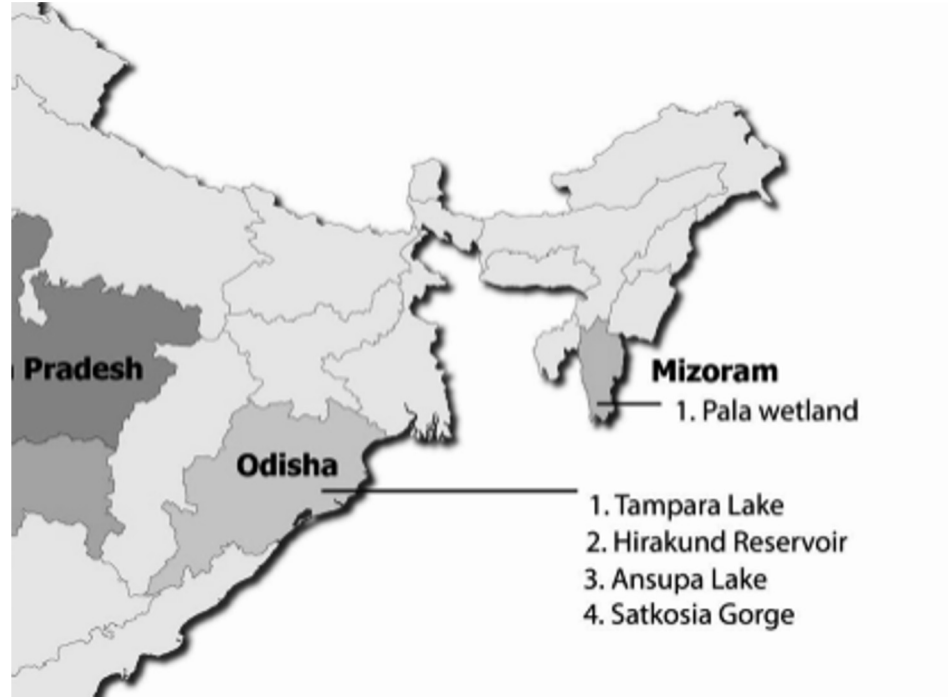
# Recently Added Ramsar Sites (Odisha)

- **Ansupa Lake (Natural wetland)**
- An oxbow lake formed by River Mahanadi and the largest freshwater lake of Odisha.
- **Hirakud Reservoir (Man-made wetland)**
- The largest earthen dam built across Mahanadi river in Odisha started operating in 1957.
- **Tampara Lake (Natural Wetland)**
- **Tamp:** The depression on the ground gradually filled with rainwater from catchment flow and was called. Hence, the name “Tampara”.



# Recently Added Ramsar Sites (Mizoram)

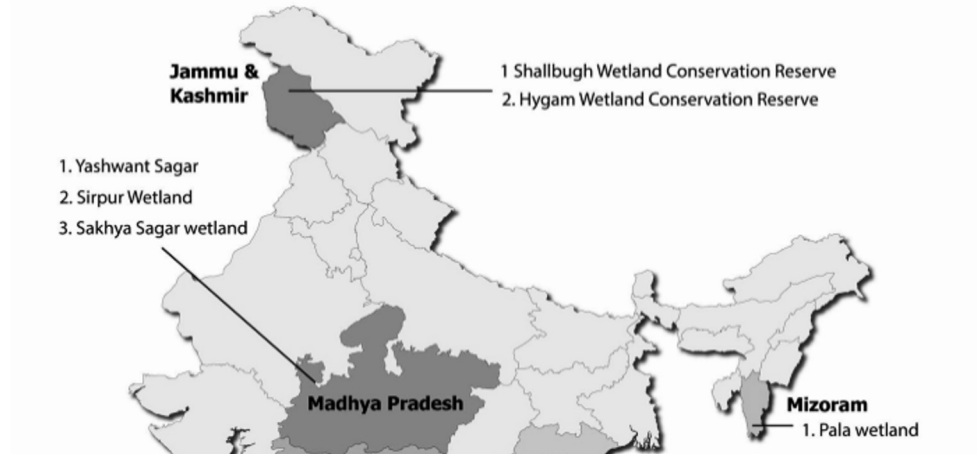
- **Pala Wetland (Natural wetland)**
- Largest natural wetland in the state of Mizoram.
- Revered by the local **Mara people**.
- **Important Species:** Sambar deer, wild pig, barking deer, Hoolock gibbon, Phayre's leaf monkey, slow loris, elongated tortoise, Asian brown tortoise, and black soft shelled turtle etc.



# Recently Added Ramsar Sites (Madhya Pradesh)



- **Sakhya Sagar**
- Created due to damming of the **Manier River** in 1918 by the Maharaja of Scindias.
- The reservoir lies within the Madhav National Park
- **Sirpur Wetland (Man-made wetland)**
- **Yashwant Sagar (Man-made wetland)**
- A dam reservoir on Gambhir river.



- **Context**
- Eight wild African Cheetahs (5 female and 3 male) from Namibia, were recently introduced in Kuno National Park (KNP), Madhya Pradesh under Project Cheetah.
- Another 12 cheetahs were brought in from South Africa recently
- Cheetah was declared extinct from India in 1952 becoming only large wild mammalian species that went extinct since independence (1947)
- **African Cheetah**
- Both a flagship species (selected to act as an ambassador, icon or symbol for a defined habitat, issue, campaign or environmental cause) and an umbrella species (its conservation ensures conservation of other species of wild animals and forests).

# Cheetah

|                                 | African Cheetah                                                                                | Asiatic Cheetah                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Pictorial Representation</b> |              |                                           |
| <b>IUCN Status</b>              | Vulnerable                                                                                     | Critically Endangered                                                                                                        |
| <b>CITES Status</b>             | Appendix I                                                                                     | Appendix I                                                                                                                   |
| <b>Distribution</b>             | Africa (Northwest Africa, East Africa, and Southern Africa)                                    | Only few left in Iran                                                                                                        |
| <b>Physical Characteristics</b> | Bigger in size as compared to Asiatic Cheetah, slightly bigger build and sturdy legs and neck. | Slightly smaller and slender than the African Cheetah. Their neck is much smaller and slender. Also, their legs are slender. |
| <b>Food Intake</b>              | Diverse diet due to bigger habitat                                                             | Limited source especially medium sized prey like Chinkara, Gazelle etc.                                                      |

# Cheetah

- Characteristics of African cheetah :
- World's fastest land mammal, capable of running at 80 to 128 km/h.
- Diurnal i.e., they hunt during the day.
- Unlike other big cats (lions, tigers, etc.) don't roar.
- Gestation period- 93 days.



- **Live in three main social groups:** females and their cubs, male coalitions, and solitary males.
- Females lead a nomadic life searching for prey in large home ranges, males are more sedentary and instead establish much smaller territories.
- Average female home ranges have been estimated to be about 750 sq km.
- Among large carnivores, conflicts with human interests are lowest for cheetahs, as they are not a threat to humans and usually do not attack large livestock.

- **Project Cheetah**
- World's first inter-continental large wild carnivore translocation project.
- Part of the centrally Sponsored Scheme- 'Project Tiger' of India.
- **Funding:** Funds from Project Tiger and Compensatory Afforestation Fund Management & Planning Authority (CAMPA)
- **Nodal Agency:** National Tiger Conservation Authority (NTCA) authorized for funding, supervision and hand holding of the project.

- Wildlife Institute of India (WII), national and international carnivore/cheetah experts/agencies for technical and knowledge support.
- Significance of Cheetah Relocation:
- Establish viable cheetah metapopulation in India
- Cheetahs can help restore open forest and grassland ecosystems in India
- Resource mobilization, contributing to global conservation efforts
- Enhance local community livelihoods, contribute to climate change mitigation goals etc

**2014**

If a wetland of international importance is brought under the ‘Montreux Record’, what does it imply?

- a) Changes in ecological character have occurred, are occurring or are likely to occur in the wetland because of human interference.
- b) The country in which the wetland is located should enact a law to prohibit any human activity within five kilometres from the edge of the wetland
- c) The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity, and therefore the cultural diversity therein should not be destroyed
- d) It is given the status of ‘World Heritage Site’

**Answer: A**

- Ramsar Sites which are potentially at risk because of technological developments, pollution or other human interference may be placed on the Montreux Record of Ramsar Sites where changes in ecological character have occurred, are occurring, or are likely to occur.

**2020**

Which of the following Protected Areas are located in Cauvery basin?

1. Nagarhole National Park
2. Papikonda National Park
3. Sathyamangalam Tiger Reserve
4. Wayanad Wildlife Sanctuary

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 3 and 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

## Solution

(C) 1, 3 and 4 only

- Papikonda NP is located near Rajamahendravaram in the Papi Hills of the East Godavari and West Godavari districts of Andhra Pradesh
- Rest other protected areas are part of Nilgiri BR which lies in the Cauvery basin

**2020**

Which one of the following protected areas is well-known for the conservation of a sub-species of the Indian swamp deer (Barasingha) that thrives well on hard ground and is exclusively graminivorous?

- (a) Kanha National Park
- (b) Manas National Park
- (c) Mudumalai Wildlife Sanctuary
- (d) Tal Chhapar Wildlife Sanctuary

## Solution

### (A) Kanha National Park

- The revival of Barasingha, an endangered and endemic cervid, belonging to the cervidae family, in the Kanha Tiger Reserve is hailed as one of the most successful and inspiring conservation projects in the world.

2020

With reference to India's Desert National Park, which of the following statements are correct?

1. It is spread over two districts.
2. There is no human habitation inside the Park.
3. It is one of the natural habitats of the Great Indian Bustard.

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

## Solution

(C) 1 and 3 only

- Desert National Park is one of the largest protected lands in India and is located in the districts of Jaisalmer and Barmer in Rajasthan.
- The park's vegetation is protected by the local communities like the Bishnois
- It is one of the most prominent habitats for the Great Indian Bustard.

**2020**

Among the following Tiger Reserves, which one has the largest area under “Critical Tiger Habitat”?

- (a) Corbett
- (b) Ranthambore
- (c) Nagarjunsagar-Srisaillam
- (d) Sunderbans

## Solution

### (C) Nagarjunsagar-Srisaillam

| S. No. | Name of Tiger Reserve                  | State             | Year of Inclusion under Project Tiger | Year of Notification/ Re-notification under Section 38V, WPA, 1972, India | Area of the core / critical tiger habitat (In Sq. Km.) | Area of the buffer / peripheral (In Sq. Km.) | Total area (In Sq.Km.) |
|--------|----------------------------------------|-------------------|---------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------|------------------------|
| 1.     | Nagarjunsagar Srisaillam Tiger Reserve | Andhra Pradesh    | 1982-1983                             | 2007                                                                      | 2,595.72                                               | 700.59                                       | 3,296.31               |
| 2.     | Kamlang Tiger Reserve                  | Arunachal Pradesh | 2016-2017                             | 2015                                                                      | 671.00                                                 | 112.00                                       | 783.00                 |
| 3.     | Namdapha Tiger Reserve                 | Arunachal Pradesh | 1982-1983                             | 1987                                                                      | 1,807.82                                               | 245.00                                       | 2,052.82               |
| 4.     | Pakke Tiger Reserve                    | Arunachal Pradesh | 1999-2000                             | 2012                                                                      | 683.45                                                 | 515.00                                       | 1,198.45               |

**2019**

Which one of the following National Parks lies completely in the temperate alpine zone?

- (a) Manas National Park
- (b) Namdapha National Park
- (c) Neora Valley National Park
- (d) Valley of Flowers National Park

## Solution

### (D) Valley of Flowers National Park

- The Alpine forests are found all along the Himalaya at altitude ranging between 2500m to 3500m.
- Valley of Flowers National Park - lies at 3352 to 3658 meters above sea level

**2019**

Which of the following are in Agasthyamala Biosphere Reserve?

- (a) Neyyar, Peppara and Shendurney Wildlife sanctuaries; and Kalakad Mundanthurai Tiger Reserve
- (b) Mudumalai, Sathyamangalam and Wayanad Wildlife Sanctuaries; and Silent Valley National Park
- (c) Kaundinya, Gundla Brahmeswaram and Papikonda Wildlife Sanctuaries; and Mukurthi National Park
- (d) Kawal and Sri Venkateswara Wildlife Sanctuaries; and NagarjunasagarSrisailem Tiger Reserve

## Solution

(A)

- Located in the Western Ghats in the south of the country, the Agasthyamala Biosphere Reserve is a unique genetic reservoir of cultivated plants. Three wildlife sanctuaries, Shendurney, Peppara, and Neyyar, are located on the site, as well as the Kalakad Mundanthurai Tiger reserve.

**2018**

In which one of the following states is Pakhui wildlife sanctuary located?

- (a) Arunachal Pradesh
- (b) Manipur
- (c) Meghalaya
- (d) Nagaland

## Solution

### (A) Arunachal Pradesh

- Pakhui Wildlife Sanctuary (862 km<sup>2</sup>, 92°36' – 93°09'E and 26°54 – 27°16'N) lies in the foothills of the Eastern Himalaya in the East Kameng District of Arunachal Pradesh.
- It was declared a sanctuary in 1977 and was earlier part of the Khellong Forest Division.
- It has been declared a tiger reserve in 2002 based on a proposal in 1999.

**2017**

(Based on Current Affairs 2016-2017)

Recently, there was a proposal to translocate some of the lions from their natural habitat in Gujarat to which one of the following sites?

- (a) Corbett National Park
- (b) Kuno Palpur Wildlife Sanctuary
- (c) Mudumalai Wildlife Sanctuary
- (d) Sariska National Park

## Solution

### (B) Kuno Palpur NP

- Kuno Palpur is now the home for cheetahs which have been reintroduced from Namibia and South Africa

**2017**

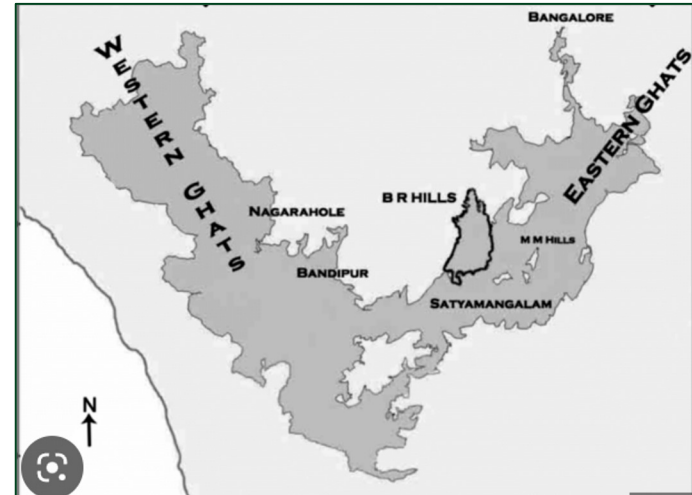
From the ecological point of view, which one of the following assumes importance in being a good link between the Eastern Ghats and the Western Ghats?

- a) Sathyamangalam Tiger Reserve
- b) Nallamala Forest
- c) Nagarhole National Park
- d) Seshachalam Biosphere Reserve

## Solution

### (A) Sathyamangalam Tiger Reserve

- Sathyamangalam Wildlife Sanctuary and Tiger Reserve is a protected area and Tiger Reserve along the Western Ghats in the Indian state of Tamil Nadu.



**2015**

Which one of the following National Parks has a climate that varies from tropical to subtropical, temperate and arctic?

- (a) Khangchendzonga National park
- (b) Nandadevi National Park
- (c) Neora Valley National Park
- (d) Namdapha National park

## Solution

### (D) Namdapha National park

- Namdapha National Park is located in Arunachal Pradesh. The climate of this area varies from tropical to subtropical, temperate and arctic.
- It is tropical and subtropical in southern regions and arctic type found in northern part of the park.

**2015**

Which of the following National Parks is unique in being a swamp with floating vegetation that supports a rich biodiversity?

- (a) Bhitarkanika National Park
- (b) Keibul Lamjao National Park
- (c) Keoladeo Ghana National park
- (d) Sultanpur National park

## Solution

### (B) Keibul Lamjao National Park

- It is the only floating National Park in the world, the Keibul Lamjao National Park located on the Loktak Lake is the last natural habitat of the 'Sangai' (*Rucervus eldii eldii*), the dancing deer of Manipur.
- Phumdis are the heterogeneous mass of vegetation, soil and organic matter at various stages of decomposition.



**2014**

Consider the following pairs

1. Dampa Tiger Reserve : Mizoram
2. Gumti Wildlife Sanctuary : Sikkim
3. Saramati Peak : Nagaland

Which of the above pairs is /are correctly matched?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

## Solution

(C) 1 and 3 only

- Dampa TR- Mizoram
- Gumti WS- Tripura
- Saramati Peak- Nagaland

**2013**

Consider the following pairs of parks and rivers passing through them:

- |                                 |        |
|---------------------------------|--------|
| 1. Corbett National Park:       | Ganga  |
| 2. Kaziranga National Park:     | Manas  |
| 3. Silent Valley National Park: | Kaveri |

Which of the above pairs is/are correctly matched?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 3 only
- d) None

## Solution

(D) None

- River Ramganga flows through the Corbett National park along with other rivers like Sonanadi, Palain, and Mandal. Hence, pair 1 is incorrect.
- Four rivers (Brahmaputra, Diphlu, Mora Diphlu, and Mora Dhansiri) flow through Kaziranga National Park. Hence, pair 2 is incorrect.
- River like Kunthi Puzha flows through Silent Valley National Park. Hence, pair 3 is incorrect.